LINEAR MINIMUM MEAN SQUARE ERROR (LMMSE) ESTIMATE

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Cerchiamo di imporre la linearità

MMSE estimator in the Gaussian case

- The MMSE estimator of vector θ, given the observed data vector **Y**, is given by the mean vector of the *a posteriori* distribution, i.e. of the pdf of θ conditioned to **Y**. The MSE matrix is obtained by averaging w.r.t. the data **Y** the conditional covariance matrix (the covariance matrix of the *a posteriori* pdf).

- When θ and **Y** are **jointly Gaussian distributed**, we found:

Non la dip. lineare

$$\hat{\boldsymbol{\theta}}_{MMSE} = E\left\{\boldsymbol{\theta}\middle|\mathbf{Y}\right\} = \boldsymbol{\eta}_{\theta|Y} = \boldsymbol{\eta}_{\theta} + \mathbf{C}_{\theta Y}\mathbf{C}_{Y}^{-1}(\mathbf{Y}-\boldsymbol{\eta}_{Y})$$

$$MSE\left\{\hat{\boldsymbol{\theta}}_{MMSE}\right\} = E\left\{(\boldsymbol{\theta}-\boldsymbol{\eta}_{\theta|Y})(\boldsymbol{\theta}-\boldsymbol{\eta}_{\theta|Y})^T\right\} = E_Y\left\{\mathbf{C}_{\theta|Y}\right\} = \mathbf{C}_{\theta|Y}$$

$$\mathbf{C}_{\theta|Y} = E\left\{(\boldsymbol{\theta}-\boldsymbol{\eta}_{\theta|Y})(\boldsymbol{\theta}-\boldsymbol{\eta}_{\theta|Y})^T\middle|\mathbf{Y}\right\} = \mathbf{C}_{\theta} - \mathbf{C}_{\theta Y}\mathbf{C}_{Y}^{-1}\mathbf{C}_{Y\theta}$$

- In this case, the **MMSE** estimator is **linear**, hence very easy to implement.

MMSE estimator in the Gaussian case

- In the case the parameter to be estimated is a scalar, i.e. θ=θ, with $\theta \in \mathcal{N}\left(\eta_\theta, \sigma_\theta^2\right)$, the previous equations become:

$$\hat{\theta}_{MMSE} = E\left\{\theta \middle| \mathbf{Y}\right\} = \eta_{\theta|Y} = \eta_\theta + \mathbf{c}_{\theta Y}\mathbf{C}_Y^{-1}(\mathbf{Y} - \boldsymbol{\eta}_Y)$$

$$MSE\left\{\hat{\theta}_{MMSE}\right\} = E\left\{(\theta - \eta_{\theta|Y})^2\right\} = E_Y\left\{\sigma_{\theta|Y}^2\right\} = \sigma_{\theta|Y}^2$$

$$\sigma_{\theta|Y}^2 = E\left\{(\theta - \eta_{\theta|Y})^2 \middle| \mathbf{Y}\right\} = \sigma_\theta^2 - \mathbf{c}_{\theta Y}\mathbf{C}_Y^{-1}\mathbf{c}_{Y\theta} = \sigma_\theta^2 - \mathbf{c}_{Y\theta}^T\mathbf{C}_Y^{-1}\mathbf{c}_{Y\theta}$$

$$\mathbf{c}_{\theta Y} \triangleq E\left\{(\theta - \eta_\theta)(\mathbf{Y} - \boldsymbol{\eta}_Y)^T\right\} = \left(E\left\{(\mathbf{Y} - \boldsymbol{\eta}_Y)(\theta - \eta_\theta)\right\}\right)^T = \left(\mathbf{c}_{Y\theta}\right)^T$$

$$\left[\mathbf{c}_{\theta Y}\right]_i = E\left\{(\theta - \eta_\theta)\left(Y_i - \eta_{Y_i}\right)\right\}, \quad i = 0,1,2,\cdots,N-1$$

MMSE estimator in the Gaussian case

- In the case also the data vector **Y** is a scalar, i.e. N=1 and **Y**=Y, we found the well-known "**regression line**":

$$\hat{\theta}_{MMSE} = E\left\{\theta \middle| Y\right\} = \eta_{\theta|Y} = \eta_\theta + c_{\theta Y}\left(\sigma_Y^2\right)^{-1}(Y-\eta_Y)$$

$$= \eta_\theta + \rho_{\theta Y}\sigma_\theta\sigma_Y\left(\sigma_Y^2\right)^{-1}(Y-\eta_Y)$$

$$= \eta_\theta + \rho_{\theta Y}\frac{\sigma_\theta}{\sigma_Y}(Y-\eta_Y)$$

→ Normalized covariance

$$MSE\left\{\hat{\theta}_{MMSE}\right\} = E\left\{(\theta-\eta_{\theta|Y})^2\right\} = E_Y\left\{\sigma_{\theta|Y}^2\right\} = \sigma_{\theta|Y}^2$$

$$\sigma_{\theta|Y}^2 = E\left\{(\theta-\eta_{\theta|Y})^2 \middle| Y\right\} = \sigma_\theta^2 - c_{\theta Y}\left(\sigma_Y^2\right)^{-1}c_{\theta Y} = \sigma_\theta^2 - \frac{\left(\rho_{\theta Y}\sigma_\theta\sigma_Y\right)^2}{\sigma_Y^2}$$

$$= \sigma_\theta^2\left(1-\rho_{\theta Y}^2\right)$$

Linear MMSE (LMMSE) estimator: the scalar case

- If the parameter vector θ to be estimated and the observed data **Y** are **<u>not</u> jointly Gaussian**, the derivation of the mean (or even of the mode) of the *a posteriori* pdf can be very cumbersome and sometimes unfeasible, at least in closed form.

- The MMSE/MAP estimators are in general **non-linea**r, so they may be difficult to implement. Sometimes implementation requires iterative numerical calculations to converge to the optimal solution.

- If it is of fundamental importance to keep **low** the **computational complexity**, a possible approach is to impose the estimator to be **linear** and to look for the coefficients of the linear combination that minimize the MSE: (No assumption, to the distribution)

→ **Linear Minimum Mean Square Error (LMMSE)**

Migliore stimatore possibile ∀ distribuzione (lineare) → Regression line

- Let us derive first the **LMMSE** in the simple case of scalar parameter θ to be estimated and scalar observed data Y:

$$\hat{\theta} = aY + b \quad \rightarrow \quad \varepsilon = \theta - \hat{\theta} = \theta - aY - b$$

Linear MMSE (LMMSE) estimator: the scalar case

- The **LMMSE** estimator is obtained by looking for the values of parameters *a* and *b* that minimize the MSE:

$$\varepsilon = \theta - \hat{\theta} = \theta - aY - b \quad \rightarrow \quad MSE\left\{\hat{\theta}\right\} = E\left\{\varepsilon^2\right\} = E\left\{\left(\theta - aY - b\right)^2\right\}$$

- The optimal solution can be found by differentiating the MSE w.r.t. *a* e *b,* setting the derivatives equal to 0 and solving the system of two linear equations in the two unknowns:

$$\begin{cases} \dfrac{\partial E\left\{\varepsilon^2\right\}}{\partial b} = 0 \\ \dfrac{\partial E\left\{\varepsilon^2\right\}}{\partial a} = 0 \end{cases}$$

Linear MMSE (LMMSE) estimator: the scalar case

$$MSE\left\{\hat{\theta}\right\} = E\left\{\varepsilon^2\right\} = E\left\{\left(\theta - aY - b\right)^2\right\}$$

$$\frac{\partial E\left\{\varepsilon^2\right\}}{\partial b} = \frac{\partial E\left\{\left(\theta - aY - b\right)^2\right\}}{\partial b} = E\left\{\frac{\partial\left\{\left(\theta - aY - b\right)^2\right\}}{\partial b}\right\} = -2E\left\{\left(\theta - aY - b\right)\right\}$$

$$= -2E\left\{\varepsilon\right\} = 0 \qquad \rightarrow \qquad \text{ubiased estimator}$$

$$\frac{\partial E\left\{\varepsilon^2\right\}}{\partial a} = \frac{\partial E\left\{\left(\theta - aY - b\right)^2\right\}}{\partial a} = E\left\{\frac{\partial\left\{\left(\theta - aY - b\right)^2\right\}}{\partial a}\right\} = E\left\{-2Y\left(\theta - aY - b\right)\right\}$$

a – How much you Relay on the data

$$= -2E\left\{Y\varepsilon\right\} = 0 \qquad \rightarrow \qquad \text{error} \perp \text{data (Ortogonality Principle)}$$

- **The LMMSE estimator is derived by imposing that the estimator is unbiased and the estimation error is orthogonal to the observed data.**

Linear MMSE (LMMSE) estimator: the scalar case

- Let us derive explicitly the values of the parameters *a* and *b* which minimize the MSE.

- From the 1st equation we get *b*:

$$E\{\theta - aY - b\} = 0 \quad \rightarrow \quad \eta_\theta - a\eta_Y - b = 0 \quad \rightarrow \quad b = \eta_\theta - a\eta_Y$$

- Then, we insert *b* in the 2nd equation and we solve w.r.t. *a*:

$$E\{Y\varepsilon\} = E\{Y(\theta - aY - b)\} = E\{Y(\theta - aY - \eta_\theta + a\eta_Y)\} = 0$$

$$E\{Y(\theta - \eta_\theta - a(Y - \eta_Y))\} = 0$$

$$E\{Y(\theta - \eta_\theta - a(Y - \eta_Y))\} - \underbrace{E\{\eta_Y(\theta - \eta_\theta - a(Y - \eta_Y))\}}_{=E\{\eta_Y \varepsilon\} = \eta_Y E\{\varepsilon\} = 0} = 0$$

Linear MMSE (LMMSE) estimator: the scalar case

$$E\left\{(Y-\eta_Y)\left(\theta-\eta_\theta-a(Y-\eta_Y)\right)\right\}=0$$

$$E\left\{(Y-\eta_Y)\left(\theta-\eta_\theta-a(Y-\eta_Y)\right)\right\}=c_{Y\theta}-a\sigma_Y^2=0$$

$$a=\frac{c_{Y\theta}}{\sigma_Y^2}=\frac{\rho_{Y\theta}\sigma_\theta\sigma_Y}{\sigma_Y^2}=\rho_{Y\theta}\frac{\sigma_\theta}{\sigma_Y} \qquad \rightarrow \qquad b=\eta_\theta-a\eta_Y=\eta_\theta-\rho_{Y\theta}\frac{\sigma_\theta}{\sigma_Y}\eta_Y$$

$$\hat{\theta}_{LMMSE}=\rho_{Y\theta}\frac{\sigma_\theta}{\sigma_Y}Y+\eta_\theta-\rho_{Y\theta}\frac{\sigma_\theta}{\sigma_Y}\eta_Y=\eta_\theta+\rho_{Y\theta}\frac{\sigma_\theta}{\sigma_Y}\left(Y-\eta_Y\right)$$

- It is worth observing that <u>the LMMSE estimator coincides with the MMSE estimator in the Gaussian case</u>. Hence, this linear estimator is the optimal MMSE estimator when the parameter to be estimated and the data are jointly Gaussian and it is the best linear unbiased estimator in all the other (non-Gaussian) cases.

Linear MMSE (LMMSE) estimator: the scalar case

$$
\begin{aligned}
MSE\left\{\hat{\theta}_{LMMSE}\right\} &= E\left\{\varepsilon^2\right\} = E\left\{\left(\theta - aY - b\right)\varepsilon\right\} = E\left\{\theta\varepsilon\right\} - aE\left\{Y\varepsilon\right\} - bE\left\{\varepsilon\right\} \\
&= E\left\{\theta\varepsilon\right\} = E\left\{\theta\varepsilon\right\} - \eta_\theta E\left\{\varepsilon\right\} = E\left\{(\theta - \eta_\theta)\varepsilon\right\} \\
&= E\left\{\left(\theta - \eta_\theta\right)\left(\theta - \eta_\theta - \rho_{Y\theta}\frac{\sigma_\theta}{\sigma_Y}\left(Y - \eta_Y\right)\right)\right\} \\
&= E\left\{\left(\theta - \eta_\theta\right)^2\right\} - \rho_{Y\theta}\frac{\sigma_\theta}{\sigma_Y}E\left\{\left(\theta - \eta_\theta\right)\left(Y - \eta_Y\right)\right\} \\
&= \sigma_\theta^2 - \rho_{Y\theta}\frac{\sigma_\theta}{\sigma_Y}\left(\rho_{Y\theta}\sigma_\theta\sigma_Y\right) \\
&= \sigma_\theta^2\left(1 - \rho_{Y\theta}^2\right)
\end{aligned}
$$

sei obbligato al lineare (Non puoi fare meglio)

In terms of estimation accuracy (Gaussiana fa cagare), il non gaussiano ti permette di usare i non lineari (sono α.) maggiori, tuttavia più efficienti)

- The MSE of the LMMSE estimator obviously coincides with that of the MMSE estimator in the Gaussian case: it is a function of the variance of the parameter θ to be estimated and of the correlation coefficient between θ and the data *Y*.

✓ poiché conosco tanto

→ Gaussian: × limitato poiché non posso migliorare lo stimatore (Meh)

Geometrical interpretation of the LMMSE estimate

- **Geometrical interpretation:** Assume a linear vector space where the element of the space are random variables and where the **scalar product** and the **norm** are defined as follows: ↳ (zero mean)

$$(Y,\theta) \triangleq \operatorname{cov}\{Y,\theta\} = E\{(Y-\eta_Y)(\theta-\eta_\theta)\} = c_{Y\theta}$$

$$\|Y\| \triangleq \sqrt{(Y,Y)} = \sqrt{E\{(Y-\eta_Y)^2\}} = \sqrt{\operatorname{var}\{Y\}} = \sigma_Y$$

$$(Y,\theta) = 0 \quad \Rightarrow \quad Y \perp \theta$$

- Let us define now the unit-norm element of the space that is collinear with *Y*:

$$I_Y \triangleq \frac{Y-\eta_Y}{\sigma_Y} = \frac{Y-\eta_Y}{\|Y\|} \quad \Rightarrow \quad \|I_Y\| = 1$$

↳ base ortonormale per il sottospazio generato dai dati

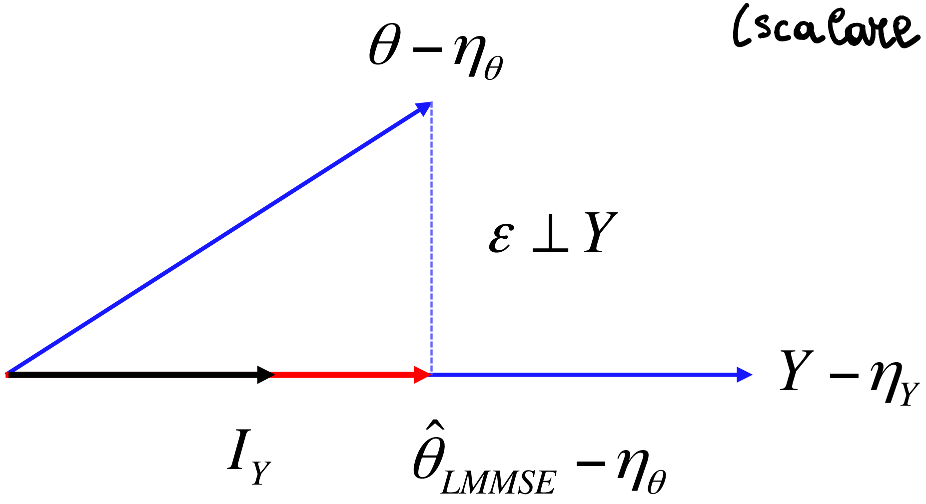
Geometrical interpretation of the LMMSE estimate

$$\hat{\theta}_{LMMSE} = \eta_\theta + \rho_{Y\theta}\frac{\sigma_\theta}{\sigma_Y}\left(Y-\eta_Y\right) \quad \Rightarrow \quad \hat{\theta}_{LMMSE} - \eta_\theta = \frac{\rho_{Y\theta}\sigma_\theta\sigma_Y}{\sigma_Y}\left(\frac{Y-\eta_Y}{\sigma_Y}\right)$$

$$\hat{\theta}_{LMMSE} - \eta_\theta = \frac{\operatorname{cov}\left\{Y,\theta\right\}}{\sigma_Y}\left(\frac{Y-\eta_Y}{\sigma_Y}\right) = \operatorname{cov}\left\{\frac{Y}{\sigma_Y},\theta\right\} I_Y = \operatorname{cov}\left\{I_Y,\theta\right\} I_Y = \underline{\left(I_Y,\theta\right)} I_Y$$

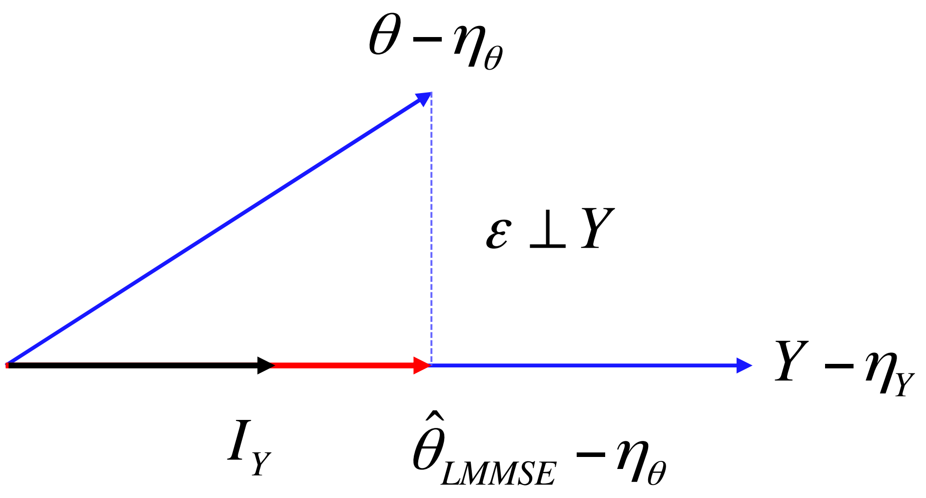
inner product
(scalare tra vettori)

- **Ortogonality Principle**: the best approximation (in the sense of minimizing the squared norm of the error) is the one which makes the error orthogonal to the data.



$$\left\|\hat{\theta}_{LMMSE} - \eta_\theta\right\| = \left\|\left(I_Y,\theta\right) I_Y\right\| = \left|\left(I_Y,\theta\right)\right| \cdot \left\|I_Y\right\| = \left|\left(I_Y,\theta\right)\right| = \left|\operatorname{cov}\left\{I_Y,\theta\right\}\right|$$

Geometrical interpretation of the LMMSE estimate



$$I_Y \triangleq \frac{Y - \eta_Y}{\sigma_Y} = \frac{Y - \eta_Y}{\|Y\|}$$

$$\left\| \hat{\theta}_{LMMSE} - \eta_\theta \right\| = \left| \operatorname{cov}\{I_Y, \theta\} \right|$$

$$\|\varepsilon\|^2 = \|\theta - \eta_\theta\|^2 - \left\| \hat{\theta}_{LMMSE} - \eta_\theta \right\|^2 = \sigma_\theta^2 - \left(\operatorname{cov}\{I_Y, \theta\} \right)^2$$

$$= \sigma_\theta^2 - \left(\operatorname{cov}\left\{ \frac{Y}{\sigma_Y}, \theta \right\} \right)^2 = \sigma_\theta^2 - \left(\frac{\operatorname{cov}\{Y, \theta\}}{\sigma_Y} \right)^2$$

$$= \sigma_\theta^2 - \left(\frac{\rho_{Y\theta} \sigma_Y \sigma_\theta}{\sigma_Y} \right)^2 = \sigma_\theta^2 - \left(\rho_{Y\theta} \sigma_\theta \right)^2 = \sigma_\theta^2 \left(1 - \rho_{Y\theta}^2 \right)$$

Example

SKIP

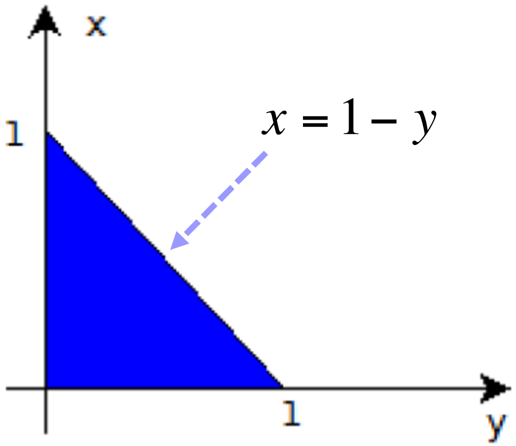
- ***Example***: The joint pdf of two random variables *X* and *Y* is given by

$$f_{YX}\left(y,x\right)=\begin{cases}6x, & 0\le x\le 1,\quad 0\le y\le 1-x\\ 0, & otherwise\end{cases}$$

- **(a)** Derive the pdf of *Y* conditioned to *X* and plot it.
- **(b)** Derive the LMMSE estimator of *Y* given the observation of *X*.
- **(c)** Derive the MMSE estimator of *Y* given the observation of *X* and its MSE.
- **(d)** Derive the MAP estimator of *Y* given the observation of *X* which is also unbiased.

Example

- The joint pdf of two random variables *X* and *Y* is different from 0 in the area of the plane reported below:



- To derive the a posteriori pdf we should apply the following relationship:

$$f_{Y|X}\left(y|x\right) = \frac{f_{YX}\left(y,x\right)}{f_X\left(x\right)}$$

- Hence, we need to derive the marginal pdf of *X*.

Example

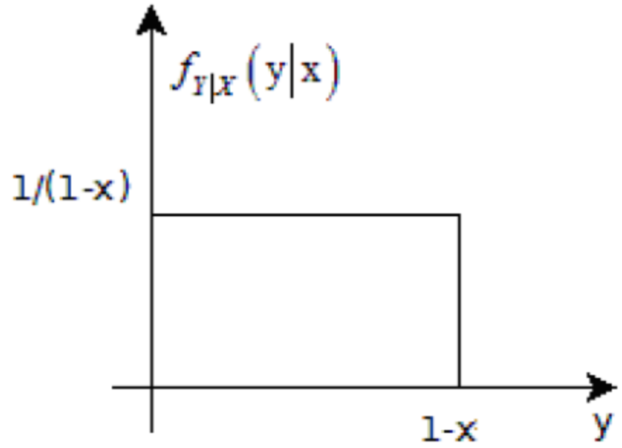
$$f_X(x) = \int_{-\infty}^{+\infty} f_{YX}(y,x)\,dy = \begin{cases} \int_0^{1-x} 6x\,dy, & 0 \le x \le 1 \\ 0, & otherwise \end{cases} = \begin{cases} 6x(1-x), & 0 \le x \le 1 \\ 0, & otherwise \end{cases}$$

$$f_{Y|X}(y|x) = \frac{f_{YX}(y,x)}{f_X(x)} = \frac{6x}{6x(1-x)} = \frac{1}{1-x}, \qquad 0 \le x \le 1, 0 \le y \le 1-x$$

$$f_{Y|X}(y|x) = \begin{cases} \dfrac{1}{1-x}, & 0 \le y \le 1-x, 0 \le x \le 1 \\ 0, & otherwise \end{cases}$$

- Hence, the r.v. *Y*/*X* is uniformly distributed in [0,1-*x*].

Example



$$Y|X \in U(0,1-x)$$

- (a) The LMMSE estimator of Y given the observation of X is given by:

$$\hat{Y}_{LMMSE} = \alpha X + \beta \quad \rightarrow \quad \varepsilon = Y - \hat{Y}_{LMMSE} = Y - \alpha X - \beta$$

$$\begin{cases} E\{\varepsilon\} = 0 \\ E\{\varepsilon X\} = 0 \end{cases}$$

Example

$$E\{Y-\alpha X-\beta\}=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(y-\alpha x-\beta)f_{XY}(x,y)\mathrm{d}y\mathrm{d}x=\int_{0}^{1}\int_{0}^{1-x}6x(y-\alpha x-\beta)\mathrm{d}y\mathrm{d}x$$

$$=\int_0^1\left[\int_0^{1-x}(6xy-6\alpha x^2-6\beta x)\mathrm{d}y\right]\mathrm{d}x$$

$$=\int_0^1\left[6x\frac{(1-x)^2}{2}-6\alpha x^2(1-x)-6\beta x(1-x)\right]\mathrm{d}x$$

$$=\int_0^1\left[3x(x^2-2x+1)-6\alpha x^2+6\alpha x^3-6\beta x+6\beta x^2\right]\mathrm{d}x$$

$$=\int_0^1\left[3x^3-6x^2+3x-6\alpha x^2+6\alpha x^3-6\beta x+6\beta x^2\right]\mathrm{d}x$$

$$=\frac{3}{4}-\frac{6}{3}+\frac{3}{2}-\frac{6}{3}\alpha+\frac{6}{4}\alpha-\frac{6}{2}\beta+\frac{6}{3}\beta$$

$$=\frac{1}{4}-\frac{1}{2}\alpha-\beta$$

Example

$$E\left\{(Y-\alpha X-\beta)X\right\}=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\left(y-\alpha x-\beta\right)xf_{XY}\left(x,y\right)\mathrm{d}y\mathrm{d}x$$

$$=\int_{0}^{1}\int_{0}^{1-x}6x^{2}\left(y-\alpha x-\beta\right)\mathrm{d}y\mathrm{d}x=\frac{1-3\alpha-5\beta}{10}$$

- By equating these two quantities to 0 we get the optimal α and β:

$$\begin{cases}\dfrac{1-2\alpha-4\beta}{4}=0\\[2ex]\dfrac{1-3\alpha-5\beta}{10}=0\end{cases}\quad\Rightarrow\quad\begin{cases}2\alpha+4\beta=1\\3\alpha+5\beta=1\end{cases}$$

$$\Rightarrow\qquad\alpha=-\frac{1}{2},\quad\beta=\frac{1}{2}$$

Example

- The LMMSE estimator is given by:

$$\hat{Y}_{LMMSE} = -\frac{1}{2}X + \frac{1}{2} = \frac{1-X}{2}$$

- **(c)** The MMSE estimator is given by:

$$\hat{Y}_{MMSE} = E\left\{Y \,\middle|\, X\right\}$$

- which is the mean value of the pdf of *Y* conditioned to *X*. Since this pdf is symmetric around (1-*x*)/2, this is also the conditional mean:

$$\hat{Y}_{MMSE} = \frac{1-X}{2} \qquad \Rightarrow \qquad \hat{Y}_{MMSE} = \hat{Y}_{LMMSE}$$

Example

Interestingly, the MMSE estimator and the LMMSE estimator coincide, even if X and Y are not jointly Gaussian.

- The MSE of the LMMSE estimator can be derived by applying the 2nd part of the Orthogonality Principle:

$$\sigma_\varepsilon^2 = E\left\{\varepsilon^2\right\} = E\left\{\varepsilon Y\right\} = E\left\{\left(Y + \frac{X}{2} - \frac{1}{2}\right)Y\right\} = E\left\{Y^2 + \frac{XY}{2} - \frac{Y}{2}\right\}$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\left(y^2 + \frac{xy - y}{2}\right)f_{XY}\left(x, y\right)\mathrm{d}y\mathrm{d}x = \int_0^1\int_0^{1-x} 6x\left(y^2 + \frac{xy - y}{2}\right)\mathrm{d}y\mathrm{d}x$$

$$= \int_0^1\int_0^{1-x}\left(6xy^2 + 3x^2y - 3xy\right)\mathrm{d}y\mathrm{d}x = \frac{1}{40}$$

Example

- **(d)** The MAP estimator is given by the mode of the a posteriori pdf:

$$\hat{Y}_{MAP} = \arg\max_{y} f_{Y|X}\left(y|x\right) = \text{ any } y \text{ in } [0,1\text{-}x]$$

- The MAP estimator is given by any value of *y* in the interval [0,1-*x*], since the pdf of *Y*|*X* is constant in this interval.

- Amongst these infinite MAP estimators, the only one that is unbiased is the one in the middle:

$$\hat{Y}_{MAP} = \frac{1-X}{2} \qquad \Rightarrow \qquad \hat{Y}_{MAP} = \hat{Y}_{MMSE} = \hat{Y}_{LMMSE}$$

LMMSE estimator: multiple observations

DA QUI NON LA CHIEDE

- Let us now derive the **LMMSE** estimator of the parameter θ in the case of multiple observed data: **Y** is an *N*-dimensional vector.

$$\hat{\theta} = \sum_{i=1}^{N} a_i Y_i + b = \mathbf{a}^T \mathbf{Y} + b, \qquad \varepsilon = \theta - \hat{\theta} = \theta - \mathbf{a}^T \mathbf{Y} - b$$

$$MSE\left\{\hat{\theta}\right\} = E\left\{\varepsilon^2\right\} = E\left\{\left(\theta - \mathbf{a}^T \mathbf{Y} - b\right)^2\right\}$$

$$\frac{\partial E\left\{\varepsilon^2\right\}}{\partial b} = \frac{\partial E\left\{\left(\theta - \mathbf{a}^T \mathbf{Y} - b\right)^2\right\}}{\partial b} = E\left\{\frac{\partial\left\{\left(\theta - \mathbf{a}^T \mathbf{Y} - b\right)^2\right\}}{\partial b}\right\} = -2E\left\{\left(\theta - \mathbf{a}^T \mathbf{Y} - b\right)\right\}$$

$$= -2E\left\{\varepsilon\right\} = 0 \qquad \rightarrow \qquad \text{unbiased estimator}$$

LMMSE estimator: multiple observations

$$E\{\varepsilon\} = 0 \quad \rightarrow \quad E\{\theta - \mathbf{a}^T\mathbf{Y} - b\} = \eta_\theta - \mathbf{a}^T\boldsymbol{\eta}_Y - b = 0 \quad \rightarrow \quad b = \eta_\theta - \mathbf{a}^T\boldsymbol{\eta}_Y$$

- Let us now insert *b* in the expression of the MSE and minimize the result w.r.t. vector **a**:

$$MSE\{\hat{\theta}\} = E\{\varepsilon^2\} = E\left\{\left(\theta - \mathbf{a}^T\mathbf{Y} - b\right)^2\right\} = E\left\{\left(\theta - \mathbf{a}^T\mathbf{Y} - \eta_\theta + \mathbf{a}^T\boldsymbol{\eta}_Y\right)^2\right\}$$

$$= E\left\{\left(\theta - \eta_\theta - \mathbf{a}^T(\mathbf{Y} - \boldsymbol{\eta}_Y)\right)^2\right\}$$

$$= E\left\{\left(\theta - \eta_\theta - \mathbf{a}^T(\mathbf{Y} - \boldsymbol{\eta}_Y)\right)\left(\theta - \eta_\theta - \mathbf{a}^T(\mathbf{Y} - \boldsymbol{\eta}_Y)\right)^T\right\}$$

$$= E\left\{\left(\theta - \eta_\theta\right)^2\right\} - \mathbf{a}^T E\left\{\left(\mathbf{Y} - \boldsymbol{\eta}_Y\right)\left(\theta - \eta_\theta\right)\right\} - E\left\{\left(\theta - \eta_\theta\right)\left(\mathbf{Y} - \boldsymbol{\eta}_Y\right)^T\right\}\mathbf{a} +$$

$$+\mathbf{a}^T E\left\{(\mathbf{Y} - \boldsymbol{\eta}_Y)(\mathbf{Y} - \boldsymbol{\eta}_Y)^T\right\}\mathbf{a}$$

$$= \sigma_\theta^2 - \mathbf{a}^T\mathbf{c}_{Y\theta} - \mathbf{c}_{\theta Y}\mathbf{a} + \mathbf{a}^T\mathbf{C}_Y\mathbf{a} = \sigma_\theta^2 - 2\mathbf{a}^T\mathbf{c}_{Y\theta} + \mathbf{a}^T\mathbf{C}_Y\mathbf{a}$$

LMMSE estimator: multiple observations

$$MSE\left\{\hat{\theta}\right\} = \sigma_\theta^2 - 2\mathbf{a}^T\mathbf{c}_{Y\theta} + \mathbf{a}^T\mathbf{C}_Y\mathbf{a}$$

- The optimal vector **a** is derived by calculating the gradient of the MSE w.r.t. **a**, setting the gradient equal to 0, and solving the equation w.r.t. **a**:

$$\nabla_{\mathbf{a}} MSE\left\{\hat{\theta}\right\} = \nabla_{\mathbf{a}}\left(\sigma_\theta^2 - 2\mathbf{a}^T\mathbf{c}_{Y\theta} + \mathbf{a}^T\mathbf{C}_Y\mathbf{a}\right) = -2\mathbf{c}_{Y\theta} + 2\mathbf{C}_Y\mathbf{a} = 0$$

$$\mathbf{a} = \mathbf{C}_Y^{-1}\mathbf{c}_{Y\theta} \qquad \rightarrow \qquad b = \eta_\theta - \mathbf{c}_{Y\theta}^T\mathbf{C}_Y^{-1}\boldsymbol{\eta}_Y$$

$$\hat{\theta}_{LMMSE} = \mathbf{c}_{Y\theta}^T\mathbf{C}_Y^{-1}\mathbf{Y} + \eta_\theta - \mathbf{c}_{Y\theta}^T\mathbf{C}_Y^{-1}\boldsymbol{\eta}_Y = \eta_\theta + \mathbf{c}_{Y\theta}^T\mathbf{C}_Y^{-1}(\mathbf{Y} - \boldsymbol{\eta}_Y)$$

- which, again, coincides with the MMSE estimator for the Gaussian case.

- The MSE of the LMMSE estimator can be obtained by inserting the optimal **a** in the MSE formula previously derived:

$$MSE\left\{\hat{\theta}_{LMMSE}\right\} = \left[\sigma_\theta^2 - 2\mathbf{a}^T\mathbf{c}_{Y\theta} + \mathbf{a}^T\mathbf{C}_Y\mathbf{a}\right]\Big|_{\mathbf{a}=\mathbf{C}_Y^{-1}\mathbf{c}_{Y\theta}}$$

$$= \sigma_\theta^2 - 2\left(\mathbf{C}_Y^{-1}\mathbf{c}_{Y\theta}\right)^T\mathbf{c}_{Y\theta} + \left(\mathbf{C}_Y^{-1}\mathbf{c}_{Y\theta}\right)^T\mathbf{C}_Y\mathbf{C}_Y^{-1}\mathbf{c}_{Y\theta}$$

$$= \sigma_\theta^2 - 2\mathbf{c}_{Y\theta}^T\mathbf{C}_Y^{-1}\mathbf{c}_{Y\theta} + \mathbf{c}_{Y\theta}^T\mathbf{C}_Y^{-1}\mathbf{C}_Y\mathbf{C}_Y^{-1}\mathbf{c}_{Y\theta}$$

$$= \sigma_\theta^2 - \mathbf{c}_{Y\theta}^T\mathbf{C}_Y^{-1}\mathbf{c}_{Y\theta}$$

→ Data uncorrelated to the parameter we want to derive.

$$MSE\left\{\hat{\theta}_{LMMSE}\right\} = \sigma_\theta^2 - \mathbf{c}_{Y\theta}^T\mathbf{C}_Y^{-1}\mathbf{c}_{Y\theta}$$

LMMSE estimator: multiple observations

- In summary, the parameter *b* is chosen in such a way that the LMMSE estimator is **unbiased**:

$A = U \Lambda U^H$

$\sqrt{A} = U \sqrt{\Lambda} U^H$

$$E\left\{\theta - \hat{\theta}_{LMMSE}\right\} = E\left\{\theta\right\} - E\left\{\eta_\theta + \mathbf{c}_{Y\theta}^T \mathbf{C}_Y^{-1}(\mathbf{Y} - \boldsymbol{\eta}_Y)\right\} = \eta_\theta - \eta_\theta = 0$$

- The vector **a** is chosen in such a way that the estimation error is orthogonal to the data (**Ortogonality Principle**): $E\left\{\varepsilon Y_i\right\} = 0, \quad i = 1, 2, \cdots, N$

$$E\left\{\varepsilon \mathbf{Y}^T\right\} = E\left\{\varepsilon \mathbf{Y}^T\right\} - E\left\{\varepsilon\right\}\boldsymbol{\eta}_Y^T = E\left\{\varepsilon(\mathbf{Y} - \boldsymbol{\eta}_Y)^T\right\}$$
$$= E\left\{\left(\theta - \eta_\theta - \mathbf{a}^T(\mathbf{Y} - \boldsymbol{\eta}_Y)\right)(\mathbf{Y} - \boldsymbol{\eta}_Y)^T\right\}$$
$$= \mathbf{c}_{\theta Y} - \mathbf{a}^T \mathbf{C}_Y = \mathbf{c}_{\theta Y} - \mathbf{c}_{Y\theta}^T \mathbf{C}_Y^{-1} \mathbf{C}_Y = \mathbf{c}_{\theta Y} - \mathbf{c}_{Y\theta}^T = 0$$

$$\text{since} \quad \mathbf{c}_{\theta Y} = (\mathbf{c}_{Y\theta})^T$$

Geometrical interpretation of the LMMSE estimate

Hai applicato Gram-Schmidt (Algebra 1)

$$\hat{\theta}_{LMMSE} = \eta_\theta + \mathbf{c}_{Y\theta}^T \mathbf{C}_Y^{-1}(\mathbf{Y} - \boldsymbol{\eta}_Y) \quad \Rightarrow \quad \hat{\theta}_{LMMSE} - \eta_\theta = \mathbf{c}_{Y\theta}^T \mathbf{C}_Y^{-1/2} \mathbf{C}_Y^{-1/2}(\mathbf{Y} - \boldsymbol{\eta}_Y)$$

$$\mathbf{I}_Y \triangleq \mathbf{C}_Y^{-1/2}(\mathbf{Y} - \boldsymbol{\eta}_Y) \quad \text{data whitening (decorrelation)}$$

$$\boldsymbol{\eta}_{I_Y} = E\left\{\mathbf{I}_Y\right\} = E\left\{\mathbf{C}_Y^{-1/2}(\mathbf{Y} - \boldsymbol{\eta}_Y)\right\} = \mathbf{C}_Y^{-1/2} E\left\{\mathbf{Y} - \boldsymbol{\eta}_Y\right\} = 0$$

$$\begin{aligned}
\mathbf{C}_{I_Y} &= E\left\{\mathbf{I}_Y \mathbf{I}_Y^T\right\} = E\left\{\mathbf{C}_Y^{-1/2}(\mathbf{Y} - \boldsymbol{\eta}_Y)(\mathbf{Y} - \boldsymbol{\eta}_Y)^T \mathbf{C}_Y^{-1/2}\right\} \\
&= \mathbf{C}_Y^{-1/2} E\left\{(\mathbf{Y} - \boldsymbol{\eta}_Y)(\mathbf{Y} - \boldsymbol{\eta}_Y)^T\right\} \mathbf{C}_Y^{-1/2} \\
&= \mathbf{C}_Y^{-1/2} \mathbf{C}_Y \mathbf{C}_Y^{-1/2} = \mathbf{C}_Y^{-1/2} \mathbf{C}_Y^{1/2} \mathbf{C}_Y^{1/2} \mathbf{C}_Y^{-1/2} \\
&= \mathbf{I} \quad [N \times N \text{ identity matrix}]
\end{aligned}$$

- Hence, the elements of vector $\mathbf{I}_Y$ are zero mean random variables, mutually uncorrelated, and with unit variance.

Geometrical interpretation of the LMMSE estimate

$$\hat{\theta}_{LMMSE} - \eta_\theta = \mathbf{c}_{Y\theta}^T \mathbf{C}_Y^{-1/2} \mathbf{C}_Y^{-1/2} (\mathbf{Y} - \boldsymbol{\eta}_Y) = \left(\mathbf{C}_Y^{-1/2} \mathbf{c}_{Y\theta}\right)^T \mathbf{C}_Y^{-1/2} (\mathbf{Y} - \boldsymbol{\eta}_Y)$$

$$= \left(\mathbf{C}_Y^{-1/2} \mathbf{c}_{Y\theta}\right)^T \mathbf{I}_Y$$

PCA

since $\mathbf{c}_{Y\theta} \triangleq E\left\{(\mathbf{Y} - \boldsymbol{\eta}_Y)(\theta - \eta_\theta)\right\}$ we have:

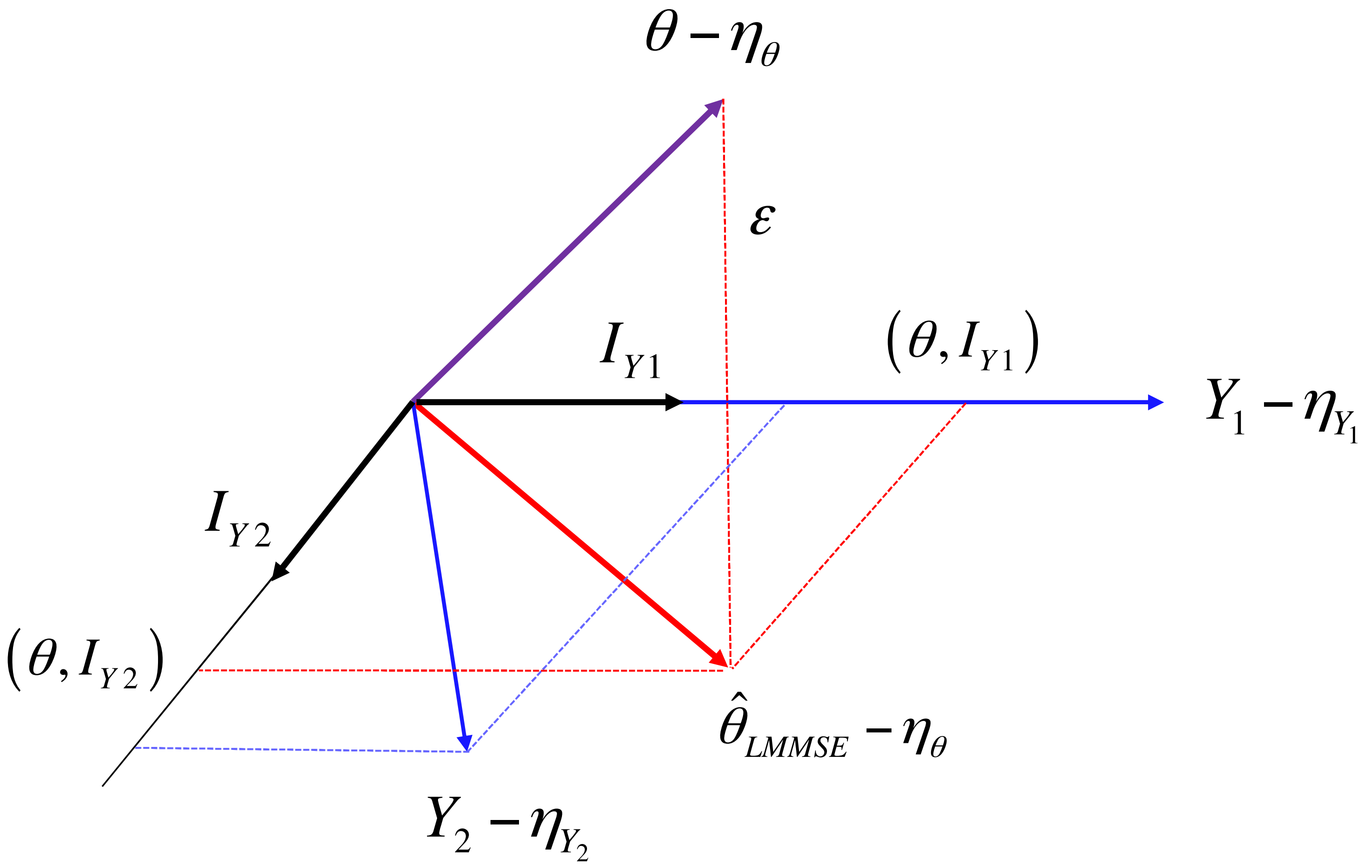
$$\mathbf{C}_Y^{-1/2} \mathbf{c}_{Y\theta} = \mathbf{C}_Y^{-1/2} E\left\{(\mathbf{Y} - \boldsymbol{\eta}_Y)(\theta - \eta_\theta)\right\} = E\left\{\mathbf{C}_Y^{-1/2} (\mathbf{Y} - \boldsymbol{\eta}_Y)(\theta - \eta_\theta)\right\}$$

$$= E\left\{\mathbf{I}_Y (\theta - \eta_\theta)\right\} = \mathbf{c}_{I_Y\theta}$$

$$\Rightarrow \qquad \hat{\theta}_{LMMSE} - \eta_\theta = \mathbf{c}_{I_Y\theta}^T \mathbf{I}_Y = \sum_{i=1}^{N} \operatorname{cov}\left\{\theta, I_{Yi}\right\} \cdot I_{Yi} = \sum_{i=1}^{N} \left(\theta, I_{Yi}\right) \cdot I_{Yi}$$

Geometrical interpretation of the LMMSE estimate

$$\hat{\theta}_{LMMSE} - \eta_\theta = \sum_{i=1}^{N} \left(\theta, I_{Y_i}\right) \cdot I_{Yi} \qquad \varepsilon \perp Y_i \quad \forall i$$



Examples

Sono dipendenti non linearmente, la dipendenza lineare ha corr. = 0

- ***Example***. We observe the following two data samples:

$$\begin{cases} X_1 = A\cos\theta + W_1 \\ X_2 = A\sin\theta + W_2 \end{cases}$$

$\gamma \to +\infty$ (W_i negligible)

Non può essere consistente, poiché ho un rapporto di non-linearità

- Find the LMMSE estimate of the phase θ, uniformly distributed in $[-\pi,\pi)$, assuming that the amplitude *A* is Rayleigh distributed and the noise samples W_1 and W_2 are zero mean correlated Gaussian random variables:

$$A \in \mathcal{R}\left(\sigma_A^2\right), \quad \theta \in \mathcal{U}(-\pi,\pi), \quad W_1 \in \mathcal{N}\left(0,\sigma_W^2\right), \quad W_2 \in \mathcal{N}\left(0,\sigma_W^2\right)$$

$$f_A(a) = \frac{2a}{\sigma_A^2} e^{-\frac{a^2}{\sigma_A^2}} u(a), \quad \text{with } E\{A\} = \sqrt{\frac{\pi}{4}}\sigma_A, \quad E\{A^2\} = \sigma_A^2$$

- **Assumption**: *A* and θ are mutually independent and are independent from the noise samples W_1 *and* W_2.

Examples

- **Assumption**: the noise samples W_1 and W_2 are partially correlated with correlation coefficient ρ.

- The LMMSE estimator is given by:

$$\hat{\theta}_{LMMSE} = \mathbf{a}^T\mathbf{X} + b = \eta_\theta + \mathbf{c}_{X\theta}^T\mathbf{C}_X^{-1}(\mathbf{X} - \boldsymbol{\eta}_X)$$

$$\eta_\theta = E\{\theta\} = \frac{1}{2\pi}\int_{-\pi}^{+\pi}\theta d\theta = 0$$

$$\boldsymbol{\eta}_X = \begin{bmatrix} E\{X_1\} \\ E\{X_2\} \end{bmatrix} = \begin{bmatrix} E\{A\}E\{\cos\theta\} + E\{W\} \\ E\{A\}E\{\sin\theta\} + E\{W\} \end{bmatrix} = \sqrt{\frac{\pi}{4}}\sigma_A \begin{bmatrix} \frac{1}{2\pi}\int_{-\pi}^{+\pi}\cos\theta d\theta \\ \frac{1}{2\pi}\int_{-\pi}^{+\pi}\sin\theta d\theta \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Examples

- Since the parameter θ and the data $\mathbf{X}$ are zero mean: $b = \eta_\theta - \mathbf{a}^T \boldsymbol{\eta}_X = 0$

$$\hat{\theta}_{LMMSE} = \mathbf{c}_{X\theta}^T \mathbf{C}_X^{-1} \mathbf{X}$$

- Coefficient vector **a** is derived by the Orthogonality Principle:

$$\mathbf{a} = \mathbf{C}_X^{-1} \mathbf{c}_{X\theta}$$

where:

$$\mathbf{c}_{X\theta} = \begin{bmatrix} E\{X_1\theta\} \\ E\{X_2\theta\} \end{bmatrix}, \quad \mathbf{C}_X = \begin{bmatrix} \mathrm{var}\{X_1\} & \mathrm{cov}\{X_1, X_2\} \\ \mathrm{cov}\{X_1, X_2\} & \mathrm{var}\{X_2\} \end{bmatrix} = \begin{bmatrix} E\{X_1^2\} & E\{X_1 X_2\} \\ E\{X_1 X_2\} & E\{X_2^2\} \end{bmatrix}$$

Examples

$$E\left\{X_1\theta\right\} = E\left\{A\theta\cos\theta + W\theta\right\} = E\left\{A\right\}E\left\{\theta\cos\theta\right\} = 0$$

$$E\left\{X_2\theta\right\} = E\left\{A\theta\sin\theta + W\theta\right\} = E\left\{A\right\}E\left\{\theta\sin\theta\right\} = \sqrt{\frac{\pi}{4}}\sigma_A$$

since:

$$E\left\{\theta\cos\theta\right\} = \frac{1}{2\pi}\int_{-\pi}^{\pi}\theta\cos\theta d\theta = \left.\frac{\theta\sin\theta}{2\pi}\right|_{-\pi}^{\pi} - \frac{1}{2\pi}\int_{-\pi}^{\pi}\sin\theta d\theta = 0$$

$$E\left\{\theta\sin\theta\right\} = \frac{1}{2\pi}\int_{-\pi}^{\pi}\theta\sin\theta d\theta = -\left.\frac{\theta\cos\theta}{2\pi}\right|_{-\pi}^{\pi} + \frac{1}{2\pi}\int_{-\pi}^{\pi}\cos\theta d\theta = 1$$

- X_1 and θ are uncorrelated!

Examples

$$E\left\{X_1^2\right\} = E\left\{(A\cos\theta + W_1)(A\cos\theta + W_1)\right\}$$

$$= E\left\{A^2\right\}E\left\{\cos^2\theta\right\} + 2E\left\{A\right\}E\left\{\cos\theta\right\}E\left\{W_1\right\} + E\left\{W_1^2\right\}$$

$$= \frac{\sigma_A^2}{2}\left(1 + E\left\{\cos(2\theta)\right\}\right) + \sigma_W^2$$

$$= \frac{\sigma_A^2}{2} + \sigma_W^2$$

$$E\left\{X_2^2\right\} = E\left\{(A\sin\theta + W_2)(A\sin\theta + W_2)\right\}$$

$$= E\left\{A^2\right\}E\left\{\sin^2\theta\right\} + 2E\left\{A\right\}E\left\{\sin\theta\right\}E\left\{W_2\right\} + E\left\{W_2^2\right\}$$

$$= \frac{\sigma_A^2}{2}\left(1 - E\left\{\cos(2\theta)\right\}\right) + \sigma_W^2$$

$$= \frac{\sigma_A^2}{2} + \sigma_W^2$$

Examples

$$
\begin{aligned}
E\left\{X_1 X_2\right\} &= E\left\{(A\cos\theta + W_1)(A\sin\theta + W_2)\right\} \\
&= E\left\{A^2\right\} E\left\{\cos\theta \sin\theta\right\} + E\left\{A\right\} E\left\{\cos\theta\right\} E\left\{W_2\right\} \\
&\quad + E\left\{A\right\} E\left\{\sin\theta\right\} E\left\{W_1\right\} + E\left\{W_1 W_2\right\} \\
&= \frac{\sigma_A^2}{2} E\left\{\sin(2\theta)\right\} + \rho\sigma_W^2 \\
&= \rho\sigma_W^2
\end{aligned}
$$

where:

$$
E\left\{A^2\right\} = \sigma_A^2, \quad E\left\{W_1 W_2\right\} = \rho\sigma_W^2, \quad E\left\{W_1\right\} = E\left\{W_2\right\} = 0
$$

Examples

- The LMMSE estimator is given by:

si potrebbe pensare che X_1 non serve poiché ho coefficiente $=0$, tuttavia poiché X_1 dipende W_1 e W_1 corr. W_2, grazie ad X_1 riesco a mitigare W_2 (è come se dentro W_2 avessi un po' di X_1

$$\hat{\theta}_{LMMSE} = \mathbf{c}_{X\theta}^T \mathbf{C}_X^{-1} \mathbf{X} = \begin{bmatrix} 0 & \sqrt{\frac{\pi}{4}}\sigma_A \end{bmatrix} \begin{bmatrix} \frac{\sigma_A^2}{2} + \sigma_W^2 & \rho\sigma_W^2 \\ \rho\sigma_W^2 & \frac{\sigma_A^2}{2} + \sigma_W^2 \end{bmatrix}^{-1} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$$

$$= \sqrt{\frac{\pi}{4}}\sigma_A \begin{bmatrix} 0 & 1 \end{bmatrix} \cdot \left(\sigma_W^2\right)^{-1} \begin{bmatrix} 1+\gamma & \rho \\ \rho & 1+\gamma \end{bmatrix}^{-1} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$$

$$= \frac{\sqrt{\frac{\pi}{4}}\sigma_A}{\sigma_W^2} \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 1+\gamma & \rho \\ \rho & 1+\gamma \end{bmatrix}^{-1} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$$

where $\gamma \triangleq \frac{\sigma_A^2}{2\sigma_W^2}$ is the Signal to Noise Ratio (SNR)

Examples

- The parameter γ is the *Signal to Noise Ratio* (*SNR*), in fact:

$$E\left\{\|\mathbf{S}\|^2\right\} = E\left\{\left(A\cos\theta\right)^2 + \left(A\sin\theta\right)^2\right\} = E\left\{A^2\right\} = \sigma_A^2$$

$$E\left\{\|\mathbf{W}\|^2\right\} = E\left\{W_1^2 + W_2^2\right\} = 2\sigma_W^2$$

$$\Rightarrow \quad SNR = \frac{E\left\{\|\mathbf{S}\|^2\right\}}{E\left\{\|\mathbf{W}\|^2\right\}} = \frac{\sigma_A^2}{2\sigma_W^2}$$

Examples

- LMMSE estimator:

$$\hat{\theta}_{LMMSE} = \sqrt{\frac{\pi}{4}} \cdot \frac{\sigma_A}{\sigma_W^2} \left((1+\gamma)^2 - \rho^2 \right)^{-1} \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 1+\gamma & -\rho \\ -\rho & 1+\gamma \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$$

$$= \sqrt{\frac{\pi}{4}} \cdot \frac{\sigma_A}{\sigma_W^2} \cdot \frac{1}{(1+\gamma)^2 - \rho^2} \begin{bmatrix} -\rho & 1+\gamma \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$$

$$= \sqrt{\frac{\pi}{4}} \cdot \frac{\sigma_A}{\sigma_W^2} \cdot \frac{1}{(1+\gamma)^2 - \rho^2} \left(-\rho X_1 + (1+\gamma) X_2 \right)$$

$$= \sqrt{\frac{\pi}{4}} \cdot \frac{\sigma_A}{\sigma_W^2} \cdot \frac{1+\gamma}{(1+\gamma)^2 - \rho^2} \left(X_2 - \frac{\rho}{1+\gamma} X_1 \right)$$

$$= \frac{\sqrt{\pi}}{\sigma_A} \cdot \frac{\gamma(1+\gamma)}{(1+\gamma)^2 - \rho^2} \left(X_2 - \frac{\rho}{1+\gamma} X_1 \right)$$

→ Sparisce solo se W_1 e W_2 uncorrelated

$$\hat{W}_{2,LMMSE} = \frac{\rho}{1+\gamma} X_1$$ (regression line)

Examples

$$\hat{\theta}_{LMMSE} = \frac{\sqrt{\pi}}{\sigma_A} \cdot \frac{\gamma(1+\gamma)}{(1+\gamma)^2 - \rho^2} \left(X_2 - \frac{\rho}{1+\gamma} X_1 \right)$$

- It's worth observing that even if X_1 and θ are uncorrelated, the LMMSE estimator makes use of the observed sample X_1, since it is correlated with X_2.
- In practice, the role of X_1 is to estimate the noise sample W_2 and remove it from the observed sample X_2, that is used to estimate θ.

$$\hat{W}_{2LMMSE} = \frac{\rho}{1+\gamma} X_1 \quad \rightarrow \quad X_2 - \frac{\rho}{1+\gamma} X_1 = X_2 - \hat{W}_{2LMMSE}$$

- In fact, since X_1 and W_2 are zero mean, we have:

$$\hat{W}_{2LMMSE} = \rho_{W_2X_1}\frac{\sigma_{W_2}}{\sigma_{X_1}}X_1 = \frac{\operatorname{cov}\{W_2, X_1\}}{\sigma_{X_1}^2}X_1$$

$$= \frac{E\{X_1W_2\}}{\frac{\sigma_A^2}{2}+\sigma_W^2}X_1 = \frac{E\{(A\cos\theta + W_1)W_2\}}{\frac{\sigma_A^2}{2}+\sigma_W^2}X_1 = \frac{E\{W_1W_2\}}{\frac{\sigma_A^2}{2}+\sigma_W^2}X_1$$

$$= \frac{\rho\sigma_W^2}{\frac{\sigma_A^2}{2}+\sigma_W^2}X_1 = \frac{\rho}{\frac{\sigma_A^2}{2\sigma_W^2}+1}X_1 = \frac{\rho}{1+\gamma}X_1$$

$$\hat{\theta}_{LMMSE} = \frac{\sqrt{\pi}}{\sigma_A}\cdot\frac{\gamma(1+\gamma)}{(1+\gamma)^2-\rho^2}\left(X_2 - \frac{\rho}{1+\gamma}X_1\right) = \frac{\sqrt{\pi}}{\sigma_A}\cdot\frac{\gamma(1+\gamma)}{(1+\gamma)^2-\rho^2}\left(X_2 - \hat{W}_{2LMMSE}\right)$$

Examples

$$\hat{\theta}_{LMMSE} = \frac{\sqrt{\pi}}{\sigma_A} \cdot \frac{\gamma(1+\gamma)}{(1+\gamma)^2 - \rho^2}\left(X_2 - \frac{\rho}{1+\gamma} X_1 \right)$$

$$\rho{=}1: \qquad \hat{\theta}_{LMMSE} = \frac{\sqrt{\pi}}{\sigma_A} \cdot \frac{1+\gamma}{2+\gamma}\left(X_2 - \frac{1}{1+\gamma} X_1 \right)$$

$$\rho{=}0: \qquad \hat{\theta}_{LMMSE} = \frac{\sqrt{\pi}}{\sigma_A} \cdot \frac{\gamma}{1+\gamma} X_2$$

- When ρ=0, X_1 is not used by the LMMSE estimator, since it is uncorrelated both with θ and W_2. Hence, it is useless for the linear estimate of θ.

- The **Mean Square Error (MSE)** of the LMMSE estimator is given by:

$$MSE\left\{\hat{\theta}_{LMMSE}\right\} = \sigma_\varepsilon^2 = \sigma_\theta^2 - \mathbf{c}_{X\theta}^T \mathbf{C}_X^{-1} \mathbf{c}_{X\theta}$$

$$\sigma_\theta^2 = E\left\{\theta^2\right\} = \frac{1}{2\pi}\int_{-\pi}^{\pi} \theta^2 d\theta = \frac{\pi^2}{3}$$

$$\sigma_\varepsilon^2 = \sigma_\theta^2 - \mathbf{c}_{X\theta}^T \mathbf{C}_X^{-1} \mathbf{c}_{X\theta} = \frac{\pi^2}{3} - \begin{bmatrix} 0 & \sqrt{\frac{\pi}{4}}\sigma_A \end{bmatrix} \begin{bmatrix} \frac{\sigma_A^2}{2} + \sigma_W^2 & \rho\sigma_W^2 \\ \rho\sigma_W^2 & \frac{\sigma_A^2}{2} + \sigma_W^2 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ \sqrt{\frac{\pi}{4}}\sigma_A \end{bmatrix}$$

$$= \frac{\pi^2}{3} - \frac{\pi\sigma_A^2}{4} \cdot \frac{1}{\sigma_W^2} \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 1+\gamma & \rho \\ \rho & 1+\gamma \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Examples

$$\sigma_\varepsilon^2 = \frac{\pi^2}{3} - \frac{\pi\sigma_A^2}{4}\cdot\frac{1}{\sigma_W^2}\begin{bmatrix}0 & 1\end{bmatrix}\begin{bmatrix}1+\gamma & \rho \\ \rho & 1+\gamma\end{bmatrix}^{-1}\begin{bmatrix}0 \\ 1\end{bmatrix}$$

$$= \frac{\pi^2}{3} - \frac{\pi}{2}\cdot\frac{\gamma}{(1+\gamma)^2-\rho^2}\begin{bmatrix}0 & 1\end{bmatrix}\begin{bmatrix}1+\gamma & -\rho \\ -\rho & 1+\gamma\end{bmatrix}\begin{bmatrix}0 \\ 1\end{bmatrix}$$

$$= \frac{\pi^2}{3} - \frac{\pi}{2}\cdot\frac{\gamma}{(1+\gamma)^2-\rho^2}\begin{bmatrix}-\rho & 1+\gamma\end{bmatrix}\begin{bmatrix}0 \\ 1\end{bmatrix}$$

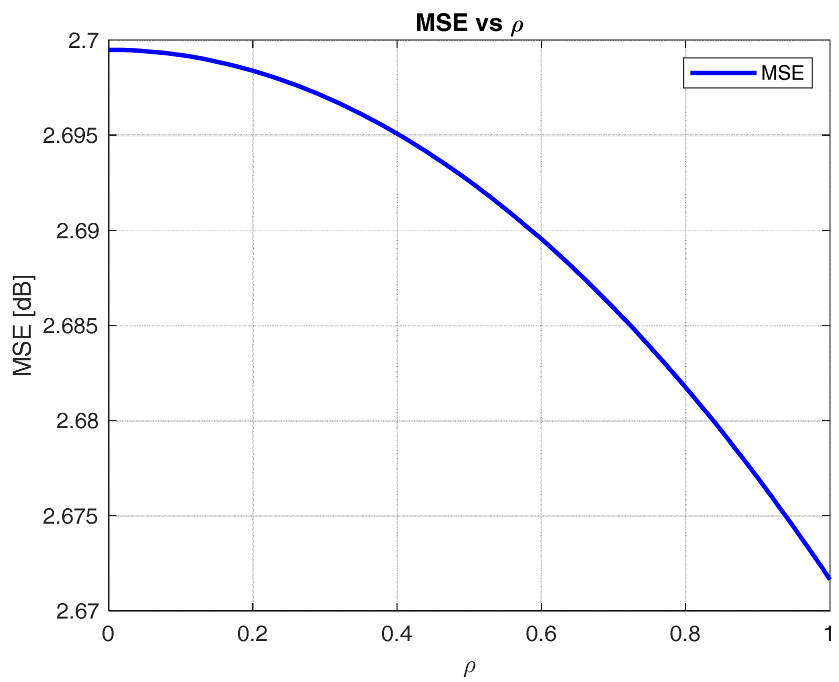
$$= \frac{\pi^2}{3} - \frac{\pi}{2}\cdot\frac{\gamma(1+\gamma)}{(1+\gamma)^2-\rho^2}$$

$$MSE\left\{\hat{\theta}_{LMMSE}\right\} = \frac{\pi^2}{3} - \frac{\pi}{2}\cdot\frac{\gamma(1+\gamma)}{(1+\gamma)^2-\rho^2}$$

Examples

$$\lim_{\gamma \to \infty} MSE\left\{\hat{\theta}_{LMMSE}\right\} = \frac{\pi^2}{3} - \frac{\pi}{2} \neq 0 \quad \Rightarrow$$ The estimator is **not consistent**.

- This is because there is a non-linear relationship between the observed data samples and the parameter to be estimated.



- The MSE decreases when ρ increases from 0 to 1.

Examples

$$\rho = 1: \qquad MSE\left\{\hat{\theta}_{LMMSE}\right\} = \frac{\pi^2}{3} - \frac{\pi}{2}\cdot\frac{1+\gamma}{2+\gamma}$$

$$\rho = 0: \qquad MSE\left\{\hat{\theta}_{LMMSE}\right\} = \frac{\pi^2}{3} - \frac{\pi}{2}\cdot\frac{\gamma}{1+\gamma}$$

- When ρ=0, X_1 is not used by the LMMSE estimator, since it is uncorrelated both with θ and W_2. Hence, it is useless for the linear estimate of θ.
- As a consequence, the MSE in the case ρ=0 is larger than in the case ρ=1.

Examples

- **Example**. The discrete observed data is a scalar (*Z*), given by signal of interest (*X*) plus noise (*N*):

$$\underbrace{Z = X}_{\text{Bernoulliana}} + N$$

- where *X* and *N* are two independent random variables with pdf's:

$$f_X(x) = \frac{1}{2}\delta(x) + \frac{1}{2}\delta(x-1), \quad f_N(n) = \frac{\lambda}{2}e^{-\lambda|n|}, \quad \text{with } \lambda > 0,\ E\{N^2\} = 2/\lambda^2$$

- We want to find the MMSE, MAP and LMMSE estimators of *X* given the observation of *Z*.

Examples

- Let us calculate first the ratio γ between the two variances:

$$\gamma \triangleq \frac{\operatorname{var}\{X\}}{\operatorname{var}\{N\}} = \frac{\sigma_X^2}{\sigma_N^2}$$

- Mean value and variance of *X:*

$$\eta_X = E\{X\} = \int_{-\infty}^{\infty} x f_X(x)dx = \int_{-\infty}^{\infty} x\left(\frac{1}{2}\delta(x) + \frac{1}{2}\delta(x-1)\right)dx = 0\cdot\frac{1}{2} + 1\cdot\frac{1}{2} = \frac{1}{2}$$

$$E\{X^2\} = \int_{-\infty}^{\infty} x^2 f_X(x)dx = \int_{-\infty}^{\infty} x^2\left(\frac{1}{2}\delta(x) + \frac{1}{2}\delta(x-1)\right)dx = \frac{1}{2}$$

$$\Rightarrow \quad \sigma_X^2 = E\{X^2\} - E^2\{X\} = 1/4$$

Examples

$Z = X + N$

$Z|X = x + \underline{N|X}$ indipendenti

$\Downarrow$

$= x + N$

- Mean value and variance of *N:*

$$\eta_N = E\{N\} = \int_{-\infty}^{+\infty} n f_N(n) dn = \int_{-\infty}^{+\infty} n \frac{\lambda}{2} e^{\lambda|n|} dn = 0$$

$$\sigma_N^2 = E\{N^2\} = \int_{-\infty}^{+\infty} n^2 f_N(n) dn = \int_{-\infty}^{+\infty} n^2 \frac{\lambda}{2} e^{\lambda|n|} dn = \frac{2}{\lambda^2}$$

- Hence, the ratio γ is given by:

$$\gamma = \frac{\sigma_X^2}{\sigma_N^2} = \frac{1/4}{2/\lambda^2} = \frac{\lambda^2}{8}$$

Examples

- The a posteriori pdf of *X* given the observed sample *Z* can be calculated by using the **Bayes' Theorem**:

$$f_{X|Z}\left(x|z\right)=\frac{f_{XZ}\left(x,z\right)}{f_Z\left(z\right)}=\frac{f_{Z|X}\left(z|x\right)f_X\left(x\right)}{f_Z\left(z\right)}$$

$$\begin{cases} f_{Z|X}\left(z|x\right)=f_N(z-x)=\dfrac{\lambda}{2}e^{-\lambda|z-x|}, \quad f_X\left(x\right)=\dfrac{1}{2}\delta\left(x\right)+\dfrac{1}{2}\delta\left(x-1\right) \\ f_{XZ}\left(x,z\right)=f_{Z|X}\left(z|x\right)f_X\left(x\right)=\dfrac{\lambda}{4}e^{-\lambda|z-x|}\left(\delta\left(x\right)+\delta\left(x-1\right)\right) \\ f_Z(z)=f_X(z)*f_N(z) \\ f_Z\left(z\right)=\displaystyle\int_{-\infty}^{\infty}f_{XZ}\left(x,z\right)\mathrm{d}x=\int_{-\infty}^{\infty}\frac{\lambda}{4}e^{-\lambda|z-x|}\left(\delta\left(x\right)+\delta\left(x-1\right)\right)\mathrm{d}x=\frac{\lambda}{4}\left(e^{-\lambda|z|}+e^{-\lambda|z-1|}\right) \end{cases}$$

marginal rule

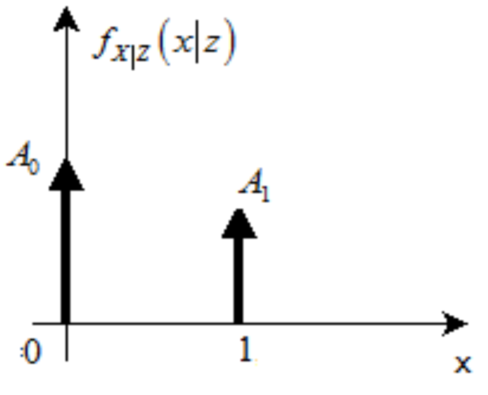
Examples

- The *a posteriori* pdf of *X* given *Z* is given by:

$$f_{X|Z}\left(x|z\right) = \frac{e^{-\lambda|z-x|}}{e^{-\lambda|z|} + e^{-\lambda|z-1|}}\left(\delta(x) + \delta(x-1)\right)$$

$$= \frac{e^{-\lambda|z|}}{e^{-\lambda|z|} + e^{-\lambda|z-1|}}\delta(x) + \frac{e^{-\lambda|z-1|}}{e^{-\lambda|z|} + e^{-\lambda|z-1|}}\delta(x-1)$$

$$= A_0(z)\cdot\delta(x) + A_1(z)\cdot\delta(x-1)$$

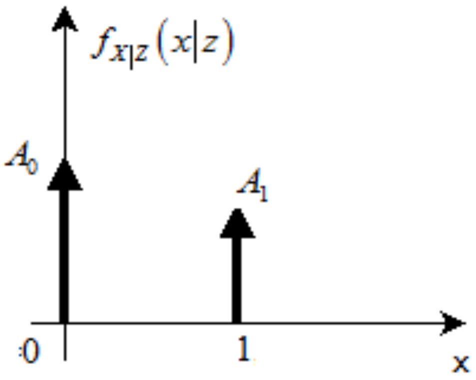


Examples

- The *a posteriori* pdf of *X* given *Z* is given by:

$$f_{X|Z}\left(x|z\right) = A_0(z)\cdot\delta(x) + A_1(z)\cdot\delta(x-1)$$

$$\begin{cases} A_0(z) = \dfrac{e^{-\lambda|z|}}{e^{-\lambda|z|} + e^{-\lambda|z-1|}} = \dfrac{1}{1+e^{-\lambda(|z-1|-|z|)}} \\ A_1(z) = \dfrac{e^{-\lambda|z-1|}}{e^{-\lambda|z|} + e^{-\lambda|z-1|}} = \dfrac{1}{1+e^{\lambda(|z-1|-|z|)}} = 1 - A_0(z) \end{cases}$$



Examples

- The MMSE is given by the mean value of the a posteriori pdf:

$$\hat{X}_{MMSE} = E\{X|Z\} = \int_{-\infty}^{\infty} x f_{X|Z}(x|z)dx$$

$$= \int_{-\infty}^{\infty} x\left[A_0(z)\cdot\delta(x) + A_1(z)\cdot\delta(x-1)\right]dx = A_1(z)$$

- Let us express the MMSE estimator in terms of γ:

Non lineare

$$\hat{X}_{MMSE} = A_1(z) = \frac{1}{1+e^{\lambda(|z-1|-|z|)}}$$

Examples

$$\hat{X}_{MMSE} = \left(1 + e^{-\lambda(|Z|-|Z-1|)}\right)^{-1}$$

$$= \left(1 + e^{-2\sqrt{2\gamma}(|Z|-|Z-1|)}\right)^{-1}$$

$$= \begin{cases} \left(1 + e^{2\sqrt{2\gamma}}\right)^{-1}, & Z < 0 \\ \left(1 + e^{-2\sqrt{2\gamma}(2Z-1)}\right)^{-1}, & 0 \le Z < 1 \\ \left(1 + e^{-2\sqrt{2\gamma}}\right)^{-1}, & Z \ge 1 \end{cases}$$

$$\gamma = \frac{\sigma_X^2}{\sigma_N^2} = \frac{\lambda^2}{8}$$

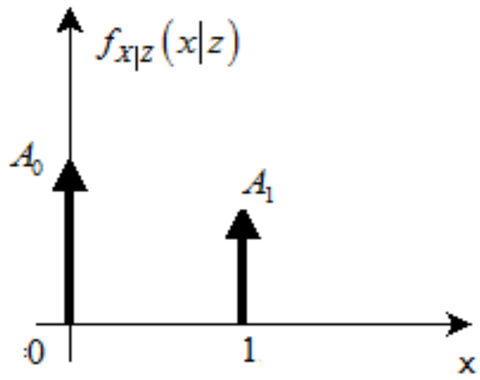
$$\Downarrow$$

$$\lambda = 2\sqrt{2\gamma}$$

- The **MMSE estimator** is a **nonlinear** function of the observed data *Z*.

Examples

- Let us now derived the **MAP estimator**.
- Since the a posteriori pdf of X given *Z* is formed by two Dirac's delta functions, the MAP estimator is given by the position of the delta with greater area.



$$\hat{X}_{MAP} = \arg\max_{x} \left\{ f_{X|Z}(x|z) \right\} = \begin{cases} 0, & \text{if } A_0(z) \geq A_1(z) \\ 1, & \text{if } A_0(z) < A_1(z) \end{cases}$$

- Let us see for which values of *z* we have $A_0 > A_1$ and vice versa.

Examples

$$A_0(z) > A_1(z) \quad \Leftrightarrow \quad \frac{1}{1+e^{-\lambda(|z-1|-|z|)}} > \frac{1}{1+e^{\lambda(|z-1|-|z|)}}$$

$$\Leftrightarrow \quad 1+e^{\lambda(|z-1|-|z|)} > 1+e^{-\lambda(|z-1|-|z|)}$$

$$\Leftrightarrow \quad \lambda\left(|z-1|-|z|\right) > -\lambda\left(|z-1|-|z|\right) \quad \Leftrightarrow \quad |z-1|-|z| > -|z-1|+|z|$$

$$\Leftrightarrow \quad |z-1| > |z| \quad \Leftrightarrow \quad z < \frac{1}{2}$$

- Hence, the **MAP estimator** can be expressed in terms of the unit step function:

$$\hat{X}_{MAP} = \begin{cases} 0, & \text{if } z \le 1/2 \\ 1, & \text{if } z > 1/2 \end{cases} \quad \Rightarrow \quad \hat{X}_{MAP} = u\left(Z - 1/2\right)$$

Examples

- The **LMMSE estimator** is given by the well-known **regression line**:

$$\hat{X}_{LMMSE} = \eta_X + \rho_{XZ} \frac{\sigma_X}{\sigma_Z} (Z - \eta_Z)$$

$$\begin{cases} \eta_X = 1/2, \quad \eta_Z = \eta_X + \eta_N = \eta_X = 1/2, \quad \sigma_X^2 = 1/4 \\ \sigma_Z^2 = \sigma_X^2 + \sigma_N^2 = \frac{1}{4} + \frac{2}{\lambda^2} = \frac{1}{4}\left(1 + \frac{1}{\gamma}\right) = \frac{1}{4} \cdot \frac{\gamma + 1}{\gamma} \\ \rho_{XZ} = \frac{E\{(X - \eta_X)(Z - \eta_Z)\}}{\sigma_X \sigma_Z} = \frac{E\{(X - \eta_X)(X + N - \eta_X)\}}{\sigma_X \sigma_Z} \\ \quad = \frac{\sigma_X^2}{\sigma_X \sigma_Z} = \frac{\sigma_X}{\sigma_Z} = \sqrt{\frac{\gamma}{\gamma + 1}} \end{cases}$$

- where we exploited the assumption that *X* and *N* are independent.

Examples

- The **LMMSE estimator** is finally obtained as:

$$\hat{X}_{LMMSE} = \eta_X + \rho_{XZ} \frac{\sigma_X}{\sigma_Z}(Z - \eta_Z)$$

$$= \frac{1}{2} + \sqrt{\frac{\gamma}{\gamma+1}}\sqrt{\frac{\gamma}{\gamma+1}}\left(Z - \frac{1}{2}\right)$$

$$= \frac{\gamma}{\gamma+1} Z + \frac{1}{2} \cdot \frac{1}{1+\gamma}$$

- Let us now investigate the behavior of the three estimators as a function of the *Signal to Noise Ratio* (SNR) γ.

Examples

- **MMSE, MAP,** and **LMMSE** estimators for γ=0:

$$\hat{X}_{MMSE} = \left(\frac{1}{1 + e^{-2\sqrt{2\gamma}\left(|Z| - |Z-1|\right)}} \right)\Bigg|_{\gamma=0} = \frac{1}{2}$$

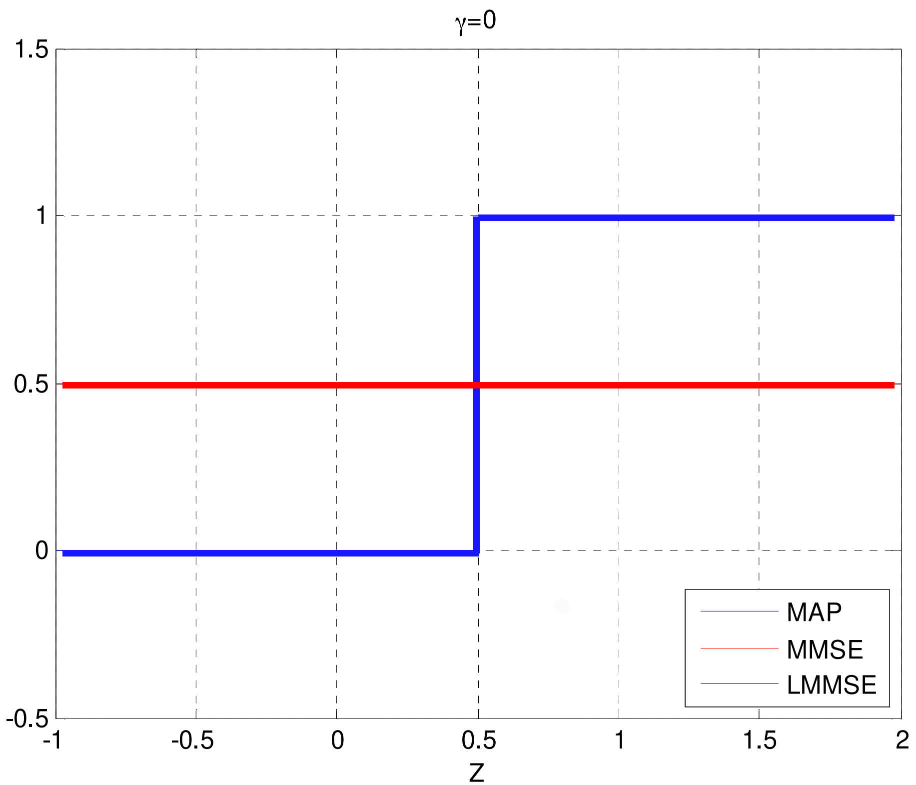
$$\hat{X}_{MAP} = u\left(Z - 1/2\right)$$

$$\hat{X}_{LMMSE} = \left(\frac{\gamma}{\gamma+1} Z + \frac{1}{2} \cdot \frac{1}{1+\gamma} \right)\Bigg|_{\gamma=0} = \frac{1}{2}$$

- When γ=0, the MMSE and LMMSE estimators coincide.
- In this case, *X* and *Z* are uncorrelated, so both the estimators estimate *X* through its mean value, which is 1/2.

Examples

- Plots of the **MMSE, MAP** and **LMMSE** estimators for γ=0:



Examples

- **MMSE, MAP** and **LMMSE** estimators for γ=1:

$$\hat{X}_{MMSE} = \left(\frac{1}{1+e^{-2\sqrt{2\gamma}\left(|Z|-|Z-1|\right)}} \right)\Bigg|_{\gamma=1} = \frac{1}{1+e^{-2\sqrt{2}\left(|Z|-|Z-1|\right)}}$$

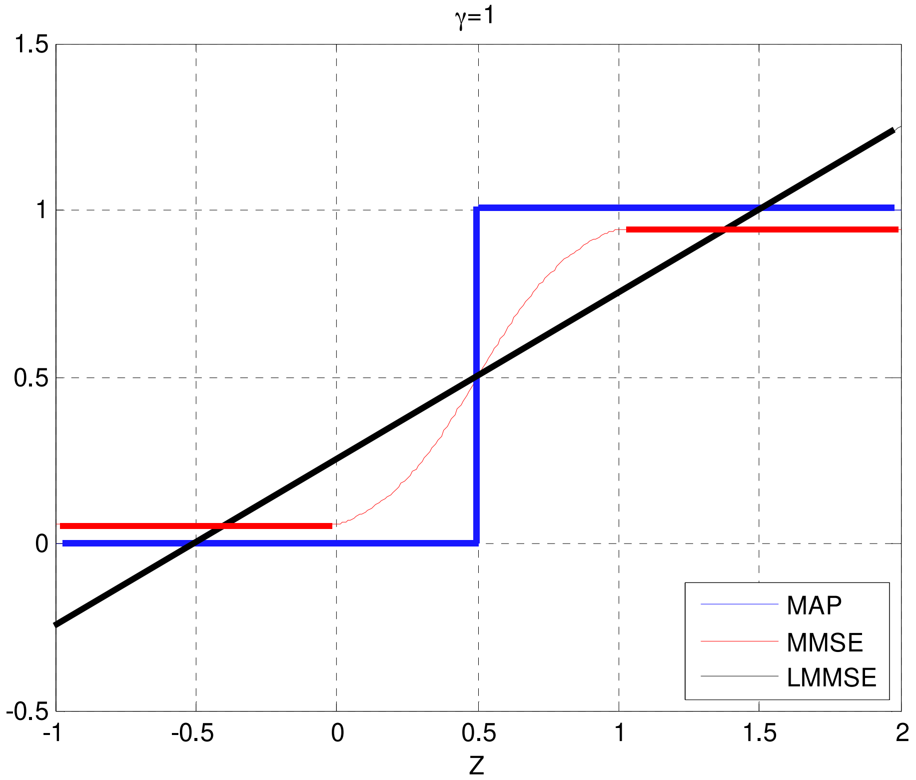
$$\hat{X}_{MAP} = u\left(Z-1/2\right)$$

$$\hat{X}_{LMMSE} = \left(\frac{\gamma}{\gamma+1} Z + \frac{1}{2} \cdot \frac{1}{1+\gamma} \right)\Bigg|_{\gamma=1} = \frac{1}{2} Z + \frac{1}{4}$$

- When γ increases the MMSE estimators tends to the MAP estimator, since the observed data *Z* tends to be more and more correlated with *X* (the noise contribution tends to be more and more negligible in the observed sample Z with respect to *X*).

Examples

- Plots of the **MMSE, MAP** and **LMMSE** estimators for γ=1:



Examples

- **MMSE, MAP** and **LMMSE** estimators for $\gamma \to \infty$:

$$\hat{X}_{MMSE} = \left. \left(\frac{1}{1 + e^{-2\sqrt{2\gamma}\left(|Z| - |Z-1|\right)}} \right) \right|_{\gamma \to \infty} = \begin{cases} 0 & se \quad Z < 1/2 \\ 1 & se \quad Z \geq 1/2 \end{cases}$$

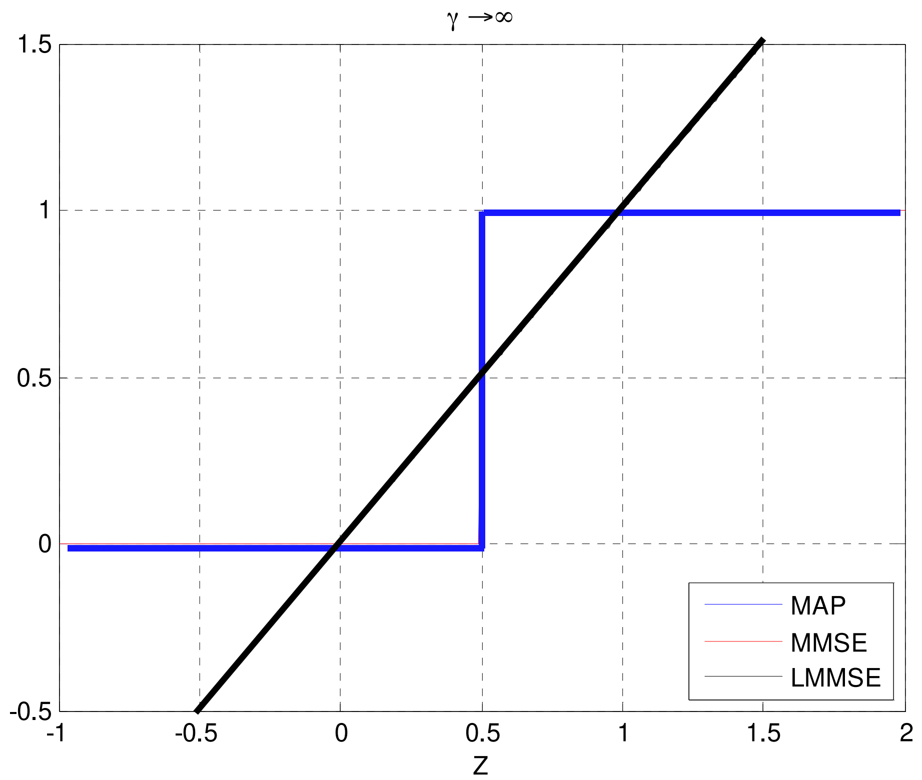
$$\hat{X}_{MAP} = u\left(Z - 1/2\right)$$

$$\hat{X}_{LMMSE} = \left. \left(\frac{\gamma}{\gamma + 1} Z + \frac{1}{2} \cdot \frac{1}{1 + \gamma} \right) \right|_{\gamma \to \infty} = Z$$

- When $\gamma \to \infty$ the MMSE estimator coincides with the MAP estimator.

Examples

- Plots of the **MMSE, MAP** and **LMMSE** estimators for $\gamma \longrightarrow \infty$:



Examples

- **Mean Square Error (MSE)** of the **LMMSE** estimator:

$$MSE\left\{\hat{X}_{LMMSE}\right\} = \sigma_X^2\left(1-\rho_{XZ}^2\right) = \sigma_X^2\left(1-\frac{\sigma_X^2}{\sigma_Z^2}\right) = \frac{1}{4}\left(1-\frac{\gamma}{\gamma+1}\right) = \frac{1}{4}\cdot\frac{1}{\gamma+1}$$

- When γ=0, *X* and *Z* are uncorrelated, so the LMMSE estimator does not use the observed data Z. In this case the MSE is maximum and it coincides with the variance of X, which is 1/4.

- When γ increases the correlation between *X* and *Z* also increases, hence the MSE decreases. The MSE tends to 0 when $\gamma\to\infty$, because of the linear relationship between Z and X and when $\gamma\to\infty$ we have $Z=X+N \to X$. The LMMSE estimator in this case is **consistent**.

Examples

- **Mean Square Error (MSE)** of the **MMSE** estimator:

$$MSE\left\{\hat{X}_{MMSE}\right\} = E_z\left\{\mathrm{var}\left\{X|Z\right\}\right\}$$

$$\mathrm{var}\left\{X|Z\right\} = E\left\{X^2|Z\right\} - \left(E\left\{X|Z\right\}\right)^2 = A_1(z) - \left(A_1(z)\right)^2$$

$$= A_1(z)\left(1 - A_1(z)\right) = A_1(z)A_0(z)$$

$$= \frac{1}{\left(1 + e^{\lambda(|z-1|-|z|)}\right)\left(1 + e^{-\lambda(|z-1|-|z|)}\right)} = \frac{1}{\left(2 + e^{\lambda(|z-1|-|z|)} + e^{-\lambda(|z-1|-|z|)}\right)}$$

$$= \frac{1}{2 + 2\cosh\left(\lambda\left(|z-1| - |z|\right)\right)}$$

$$MSE\left\{\hat{X}_{MMSE}\right\} = \int_{-\infty}^{+\infty} \frac{1}{2 + 2\cosh\left(\lambda\left(|z-1| - |z|\right)\right)} f_Z(z)dz$$

- which cannot be derived in closed form, but only via numerical integration.

Examples

- **Example**. Let us consider the following problem of **linear interpolation**. It is given a wide-sense stationary (WSS) process $X(t)$ obtained by filtering a white noise process $W(t)$ with a Moving Window Integrator (MWI):

$$X(t) = \frac{1}{T}\int_{t-T}^{t} W(\alpha)d\alpha, \qquad \text{where } S_W(f) = \frac{N_0}{2}$$

- We observe the output process at the two time instants t=0 and t=T.
- Determine the **LMMSE interpolator** that estimates $X(t)$ given the observed data $X(0)$ and $X(T)$.
- Find the MSE of the LMMSE interpolator and plot it as a function of time t.

Examples

- The WSS process $X(t)$ is obtained by filtering the process $W(t)$ through a Linear Time-Invariant (LTI) filter, a *Moving Window Integrator* (MWI):

$$X(t) = W(t) \otimes h(t)$$

- whose impulse response is given by:

$$h(t) = X(t) = \frac{1}{T}\int_{t-T}^{t} \delta(\alpha)d\alpha = \frac{1}{T}rect\left(\frac{t - T/2}{T}\right)$$

- The **LMMSE** interpolator is given by:

$$\hat{X}_{LMMSE}(t) = a_1 X(0) + a_2 X(T) + b$$

Examples

- Parameter *b* is obtained as usual by imposing that the estimator is unbiased:

$$E\left\{\varepsilon(t)\right\} = E\left\{X(t) - \hat{X}_{LMMSE}(t)\right\} = E\left\{X(t) - a_1 X(0) - a_2 X(T) - b\right\} = 0$$

$$E\left\{X(t)\right\} - a_1 E\left\{X(0)\right\} - a_2 E\left\{X(T)\right\} - b = 0$$

$$b = \eta_X(t) - a_1 \eta_X(0) - a_2 \eta_X(T) = 0$$

- Where we used the fact that:

$$\eta_X(t) = h(t) \otimes \eta_W(t) = 0 \quad [W(t) \text{ is zero mean}]$$

Examples

- Hence, the **LMMSE** estimator becomes:

$$\hat{X}_{LMMSE}(t) = a_1 X(0) + a_2 X(T)$$

- Parameters a_1 and a_2 are derived by imposing the **Orthogonality Principle**, i.e. that the estimation error is orthogonal to the data $X(0)$ and $X(T)$:

$$\begin{cases} E\{\varepsilon(t)X(0)\} = E\{[X(t) - a_1 X(0) - a_2 X(T)]X(0)\} = 0 \\ E\{\varepsilon(t)X(T)\} = E\{[X(t) - a_1 X(0) - a_2 X(T)]X(T)\} = 0 \end{cases}$$

Examples

$$\begin{cases} E\{[X(t)-a_1X(0)-a_2X(T)]X(0)\} = R_X(t)-a_1R_X(0)-a_2R_X(T)=0 \\ \\ E\{[X(t)-a_1X(0)-a_2X(T)]X(T)\} = R_X(t-T)-a_1R_X(T)-a_2R_X(0)=0 \end{cases}$$

- This system of two linear equations can be expressed as:

$$\begin{bmatrix} R_X(t) \\ R_X(t-T) \end{bmatrix} = \begin{bmatrix} R_X(0) & R_X(T) \\ R_X(T) & R_X(0) \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$$

$$\Downarrow$$

$$\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} R_X(0) & R_X(T) \\ R_X(T) & R_X(0) \end{bmatrix}^{-1} \begin{bmatrix} R_X(t) \\ R_X(t-T) \end{bmatrix}$$

Examples

- The solution is easily obtained:

$$\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \frac{1}{R_X^2(0) - R_X^2(T)} \begin{bmatrix} R_X(0) & -R_X(T) \\ -R_X(T) & R_X(0) \end{bmatrix} \begin{bmatrix} R_X(t) \\ R_X(t-T) \end{bmatrix}$$

$$= \begin{bmatrix} \dfrac{R_X(0)R_X(t) - R_X(T)R_X(t-T)}{R_X^2(0) - R_X^2(T)} \\ \dfrac{R_X(0)R_X(t-T) - R_X(T)R_X(t)}{R_X^2(0) - R_X^2(T)} \end{bmatrix}$$

- At this point, we have to calculate explicitely the Autocorrelation Function (ACF) of $X(t)$ and insert it in the above equations.
- $X(t)$ is the process obtained by linear filtering the white process $W(t)$, the LTI filter has impulse response $h(t)$ known.

Examples

- Hence, we get:

$$R_X(\tau) = R_W(\tau) \otimes h(\tau) \otimes h(-\tau) = R_W(\tau) \otimes R_h(\tau)$$

$$R_W(\tau) = \frac{N_0}{2}\delta(\tau)$$

$$R_h(\tau) = \frac{1}{T} rect\left(\frac{\tau - T/2}{T}\right) \otimes \frac{1}{T} rect\left(\frac{-\tau - T/2}{T}\right) = \frac{1}{T}\left(1 - \frac{|\tau|}{T}\right) rect\left(\frac{\tau}{2T}\right)$$

$$\Rightarrow \quad R_X(\tau) = \frac{N_0}{2T}\left(1 - \frac{|\tau|}{T}\right) rect\left(\frac{\tau}{2T}\right) \quad \Rightarrow \quad R_X(0) = \frac{N_0}{2T}, \quad R_X(T) = 0$$

- Inserting these two values of the ACF in the previous equations, we obtain the optimal parameters a_1 and a_2 (that are functions of time t):

$$\begin{cases} a_1 = \dfrac{R_X(t)}{R_X(0)} = \left(1 - \dfrac{|t|}{T}\right) rect\left(\dfrac{t}{2T}\right) \\[2em] a_2 = \dfrac{R_X(t-T)}{R_X(0)} = \left(1 - \dfrac{|t-T|}{T}\right) rect\left(\dfrac{t-T}{2T}\right) \end{cases}$$

$$\hat{X}_{LMMSE}(t) = \left(1 - \frac{|t|}{T}\right) rect\left(\frac{t}{2T}\right) X(0) + \left(1 - \frac{|t-T|}{T}\right) rect\left(\frac{t-T}{2T}\right) X(T)$$

Examples

$$\hat{X}_{LMMSE}(t) = \left(1 - \frac{|t|}{T}\right) rect\left(\frac{t}{2T}\right) X(0) + \left(1 - \frac{|t-T|}{T}\right) rect\left(\frac{t-T}{2T}\right) X(T)$$

- For 0<*t*<*T* we have:

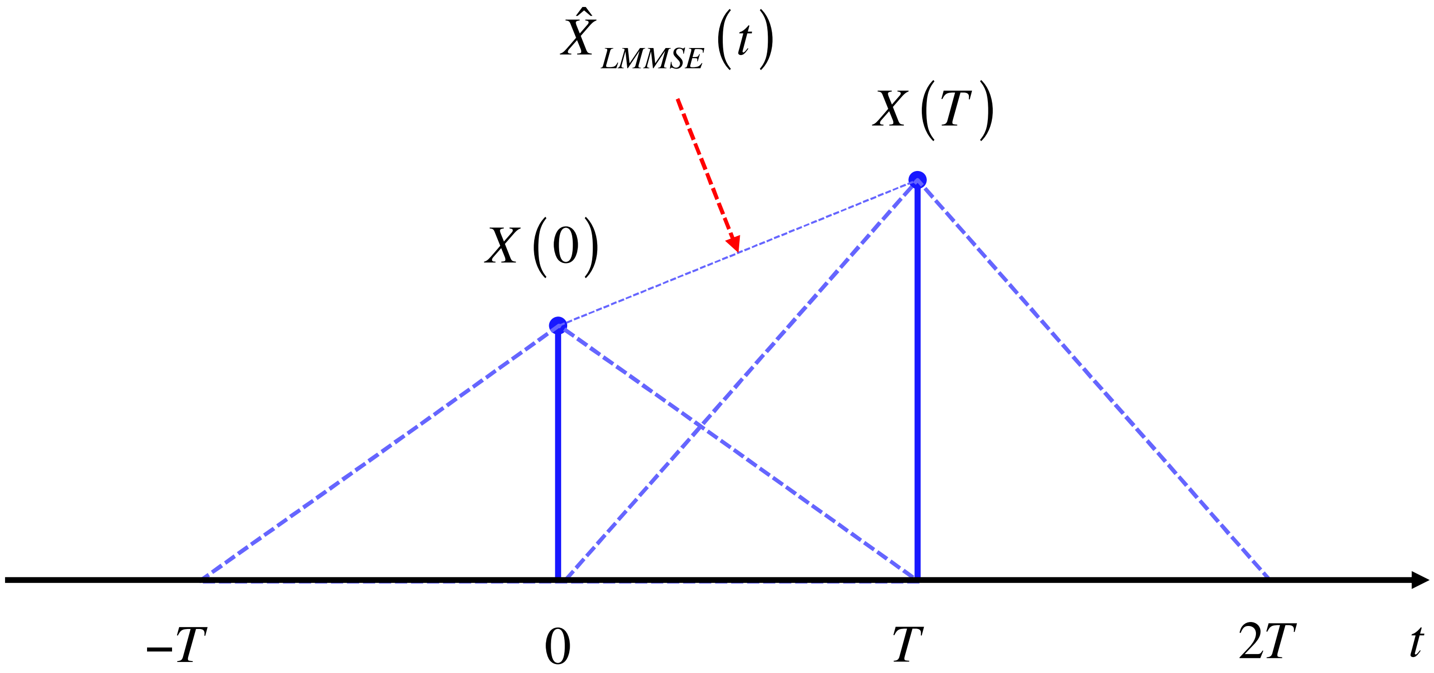
$$a_1 = 1 - \frac{t}{T} \quad \text{e} \quad a_2 = \frac{t}{T}$$

$$\Rightarrow \quad \hat{X}_{LMMSE}(t) = \left(1 - \frac{t}{T}\right) X(0) + \frac{t}{T} X(T) = X(0) + \left[X(T) - X(0)\right]\frac{t}{T}$$

- That is exactly the expression of the classical linear interpolator, in fact it is a straight line that passes through the points *X*(0) and X(*T*).

Examples

$$\hat{X}_{LMMSE}(t) = \left(1 - \frac{|t|}{T}\right) rect\left(\frac{t}{2T}\right) X(0) + \left(1 - \frac{|t-T|}{T}\right) rect\left(\frac{t-T}{2T}\right) X(T)$$



Examples

- MSE of the LMMSE interpolator:

$$MSE\left\{\hat{X}_{LMMSE}\left(t\right)\right\} = E\left\{\varepsilon^2(t)\right\} = E\left\{\varepsilon(t)\left(X(t) - \hat{X}_{LMMSE}\left(t\right)\right)\right\}$$

$$= E\left\{\varepsilon(t)X(t)\right\}$$

$$= E\left\{\left[X(t) - a_1 X(0) - a_2 X(T)\right]X(t)\right\}$$

$$= R_X(0) - a_1 R_X(t) - a_2 R_X(t-T)$$

- Inserting the expressions of the optimal a_1 and a_2 previously derived, we get the following expression for the MSE.

Examples

$$MSE\left\{\hat{X}_{LMMSE}(t)\right\} = R_X(0) - a_1 R_X(t) - a_2 R_X(t-T)$$

$$= R_X(0) - \frac{R_X^2(t)}{R_X(0)} - \frac{R_X^2(t-T)}{R_X(0)}$$

$$= R_X(0)\left[1 - \left(\frac{R_X(t)}{R_X(0)}\right)^2 - \left(\frac{R_X(t-T)}{R_X(0)}\right)^2\right]$$

$$= \frac{N_0}{2T}\left[1 - \left(1 - \frac{|t|}{T}\right)^2 rect\left(\frac{t}{2T}\right) - \left(1 - \frac{|t-T|}{T}\right)^2 rect\left(\frac{t-T}{2T}\right)\right]$$

Examples

- The expression of the MSE for *0<t<T* is given by:

$$MSE\left\{\hat{X}_{LMMSE}(t)\right\}=\frac{N_0}{2T}\left[1-\left(1-\frac{t}{T}\right)^2-\left(1-\frac{T-t}{T}\right)^2\right]=\frac{N_0}{T^2}\left(1-\frac{t}{T}\right)t,\quad t\in[0,T]$$

- MSE for *T<t<2T*:

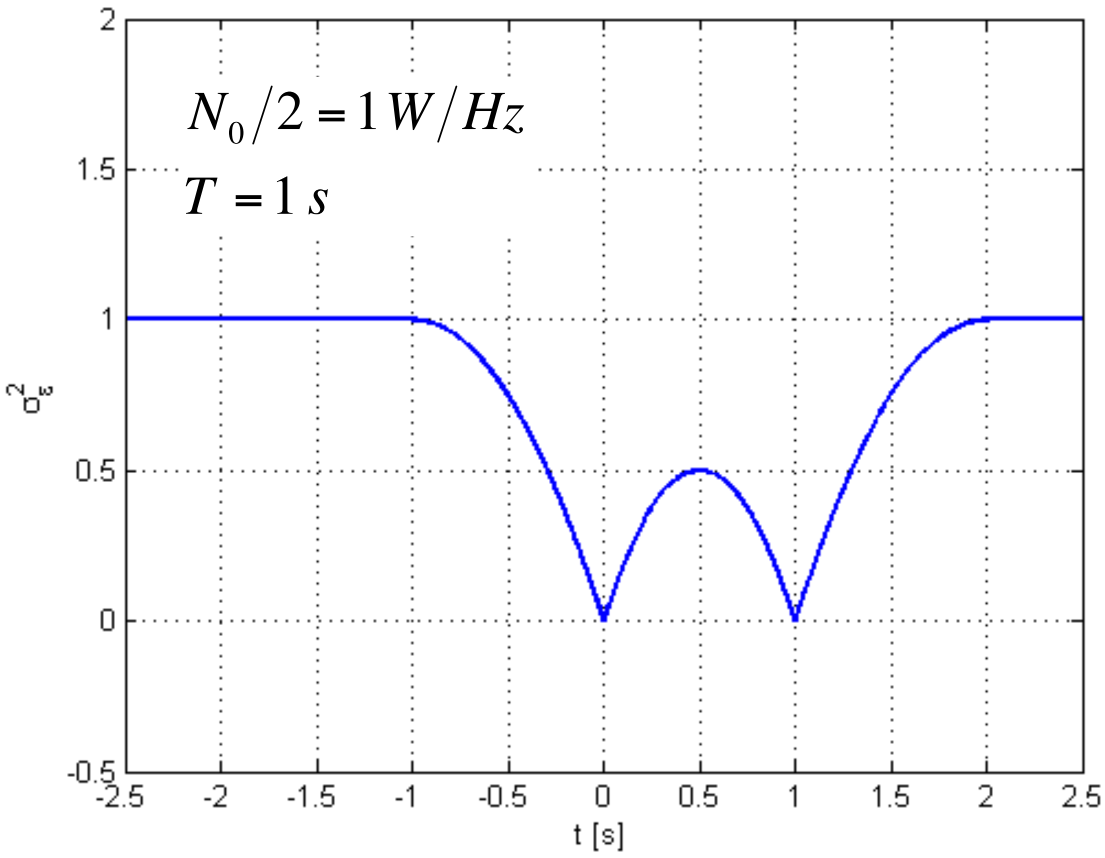
$$MSE\left\{\hat{X}_{LMMSE}(t)\right\}=\frac{N_0}{2T}\left[1-\left(1-\frac{t-T}{T}\right)^2\right]=\frac{N_0}{2T}\left[1-\left(2-\frac{t}{T}\right)^2\right],\quad t\in[T,2T]$$

- MSE for 2*T<t*:

$$MSE\left\{\hat{X}_{LMMSE}(t)\right\}=\frac{N_0}{2T},\quad t\in[2T,+\infty)$$

- The MSE is symmetric around *t*=*T*/2.

Examples



$N_0/2 = 1\,W/Hz$
$T = 1\,s$
2
1.5
1
0.5
0
-0.5
σ_ε^2
-2.5
-2
-1.5
-1
-0.5
0
0.5
1
1.5
2
2.5
t [s]

Examples

- Given the time instant t, the LMMSE estimator gives more weight to the closest sample, since it is the most correlated with the value to be estimated.
- If the time instant t is farther than T from a sample, then the weight is 0, since $X(t)$ and the data sample are uncorrelated. Remember that the correlation length for process $X(t)$ is T.
- If t=0 or t=T, the estimate is exact, since the value is known. So the MSE is 0. Within the interval $[0, T]$ the maximum MSE is for t=T/2, and this MSE is $N_0 T/4=1/2$.
- Outside the interval $[0, T]$, the MSE increases when we get further from the two samples. The maximum value for the MSE is obtained when the data are uncorrelated with the quantity to be estimated. In this case the LMMSE is given by the mean value of $X(t)$, that is 0, and the MSE is equal to the variance of $X(t)$, that is $R_X(0)=N_0/(2T)=1$.

Vector LMMSE estimator

- The **LMMSE estimator** of the random vector θ is a simple generalization of the scalar one, and obviously coincides with the MMSE estimator for the Gaussian case:

$$\begin{cases} \hat{\boldsymbol{\theta}}_{LMMSE} = \boldsymbol{\eta}_\theta + \mathbf{C}_{Y\theta}^T \mathbf{C}_Y^{-1} \left(\mathbf{Y} - \boldsymbol{\eta}_Y \right) \\ MSE\left\{ \hat{\boldsymbol{\theta}}_{LMMSE} \right\} = \mathbf{C}_\theta - \mathbf{C}_{Y\theta}^T \mathbf{C}_Y^{-1} \mathbf{C}_{Y\theta} \end{cases}$$

- It is important to stress that to implement the **LMMSE** estimator of θ it is not necessary to know the joint pdf of θ and **Y**, but we need only to know some 1st and 2nd order statistics.

Vector LMMSE estimator

$$\boldsymbol{\eta}_\theta = E\{\boldsymbol{\theta}\}$$

$$\underset{p\times p}{\mathbf{C}_\theta} = E\left\{(\boldsymbol{\theta}-\boldsymbol{\eta}_\theta)(\boldsymbol{\theta}-\boldsymbol{\eta}_\theta)^T\right\}$$

- Mean vector and covariance matrix of the vector θ we want to estimate.

$$\underset{N\times p}{\mathbf{C}_{Y\theta}} = E\left\{\left(\mathbf{Y}-\boldsymbol{\eta}_Y\right)\left(\boldsymbol{\theta}-\boldsymbol{\eta}_\theta\right)^T\right\}$$

- Cross-covariance matrix between the data vector **Y** and the vector θ we want to estimate.

$$\boldsymbol{\eta}_Y = E\{\mathbf{Y}\}$$

$$\underset{N\times N}{\mathbf{C}_Y} = E\left\{\left(\mathbf{Y}-\boldsymbol{\eta}_Y\right)\left(\mathbf{Y}-\boldsymbol{\eta}_Y\right)^T\right\}$$

- Mean vector and covariance matrix of the data vector **Y** → $\mathbf{C}_Y$ is necessary for the data *whitening*.

LMMSE estimator for the Bayesian Linear Model

- The **LMMSE** estimator in the case the observed data are represented by the **Bayesian Linear Model**, i.e. θ is random, but not necessarily Gaussian distributed, embedded in additive noise (AWN), not necessarily Gaussian distributed:

$$\underset{N\times 1}{\mathbf{X}} = \mathbf{H}\boldsymbol{\theta} + \mathbf{W} = \sum_{i=1}^{p} \theta_i \mathbf{h}_i + \mathbf{W}$$

$$\underset{N\times p}{\mathbf{H}} = \begin{bmatrix} \mathbf{h}_1 & \mathbf{h}_2 & \cdots & \mathbf{h}_p \end{bmatrix}, \quad \underset{p\times 1}{\boldsymbol{\theta}} = \begin{bmatrix} \theta_1 \\ \theta_2 \\ \vdots \\ \theta_p \end{bmatrix}$$

$$\boldsymbol{\eta}_W = E\left\{\mathbf{W}\right\} = \mathbf{0}, \quad \mathbf{C}_W = E\left\{\mathbf{W}\mathbf{W}^T\right\}$$

$$\boldsymbol{\eta}_\theta = E\left\{\boldsymbol{\theta}\right\}$$

$$\mathbf{C}_\theta = E\left\{(\boldsymbol{\theta} - \boldsymbol{\eta}_\theta)(\boldsymbol{\theta} - \boldsymbol{\eta}_\theta)^T\right\}$$

$f_{\mathbf{X},\boldsymbol{\theta}}(\mathbf{x},\boldsymbol{\theta})$ can be any, possibly unknown. $\boldsymbol{\theta}$ and $\mathbf{W}$ are uncorrelated.

LMMSE estimator for the Bayesian Linear Model

$$\boldsymbol{\eta}_X = E\{\mathbf{X}\} = \mathbf{H}\boldsymbol{\eta}_\theta$$

$$\mathbf{C}_X = E\left\{(\mathbf{X}-\boldsymbol{\eta}_X)(\mathbf{X}-\boldsymbol{\eta}_X)^T\right\} = \mathbf{H}\mathbf{C}_\theta\mathbf{H}^T + \mathbf{C}_W$$

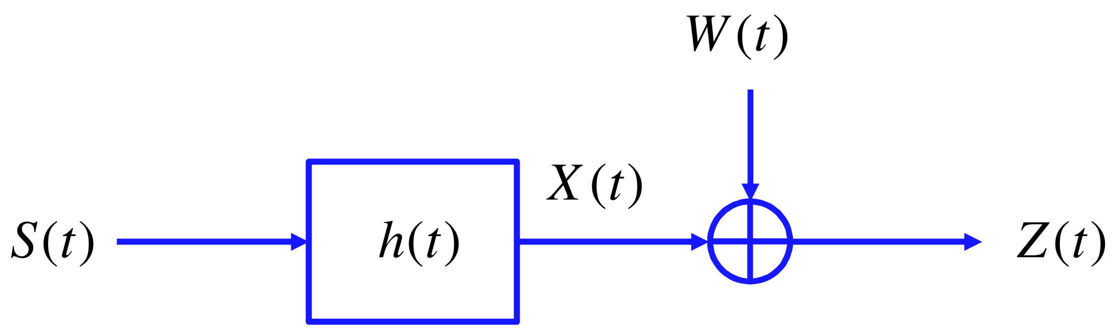
$$\mathbf{C}_{X\theta} = E\left\{(\mathbf{X}-\boldsymbol{\eta}_X)(\boldsymbol{\theta}-\boldsymbol{\eta}_\theta)^T\right\} = \mathbf{H}\mathbf{C}_\theta$$

- Inserting these relations in the general expressions of the LMMSE estimator, we get the **LMMSE estimator** for the **Bayesian Linear model** (θ is random, but not necessarily Gaussian distributed):

$$\begin{cases} \hat{\boldsymbol{\theta}}_{LMMSE} = \boldsymbol{\eta}_\theta + \mathbf{C}_\theta\mathbf{H}^T\left(\mathbf{H}\mathbf{C}_\theta\mathbf{H}^T + \mathbf{C}_W\right)^{-1}\left(\mathbf{X}-\mathbf{H}\boldsymbol{\eta}_\theta\right) \\ MSE\left\{\hat{\boldsymbol{\theta}}_{LMMSE}\right\} = \mathbf{C}_\varepsilon = \left(\mathbf{C}_\theta^{-1} + \mathbf{H}^T\mathbf{C}_W^{-1}\mathbf{H}\right)^{-1} \end{cases}$$

- "**Batch**" estimate (not recursive): we process one block of data at time.

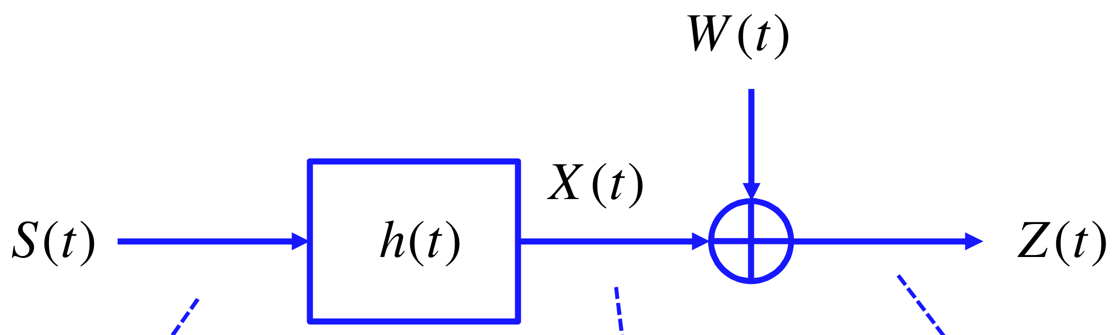
- **Example:** Problem of **linear deconvolution** of a Wide-Sense Stationary (WSS) random signal $S(t)$, when we observe a version of the signal distorted by a linear time-invariant (LTI) channel having known impulse response $h(t)$, in the presence of additive white noise $W(t)$ uncorrelated with the signal $S(t)$.

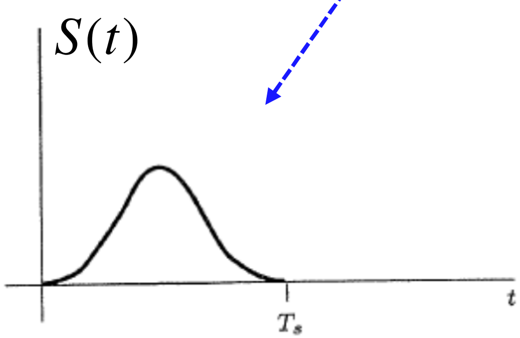


- **Problem:** retrieve (estimate) $S(t)$, given the observation $Z(t)$ in the interval $[0, T)$.

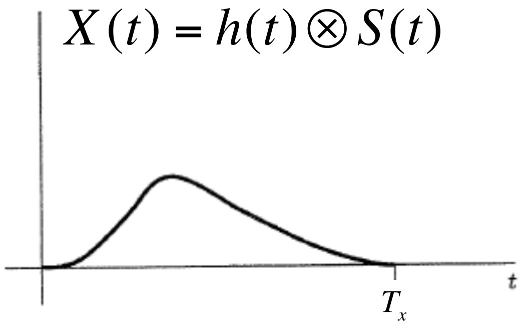
- **Assumption**: the random signal $S(t)$ is bandlimited with bandwidth B_s (or essential bandwidth at level α<<1 equal to B_s).

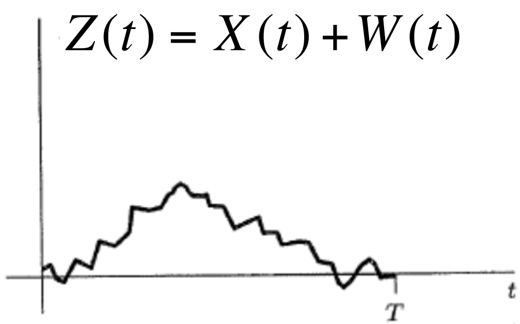
Example: Linear deconvolution





$$T_s \propto \frac{1}{B_s}$$





$$T \geq T_x = T_s + T_h$$

Example: Linear deconvolution

- To represent the received signal $Z(t)$ in discrete format, first we filter the received signal with an *anti-aliasing* filter $h_{LP}(t)$, low-pass with bandwidth $B \geq B_s$, such that $2BT >> 1$, and then we sample the received signal in the interval $[0, T)$ with sampling frequency $F_c = 2B$, i.e. sampling interval $T_c = 1/(2B)$, collecting $N = 2BT$ samples.

- $$Z(t) \otimes h_{LP}(t) = \left[X(t) + W(t) \right] \otimes h_{LP}(t) = X(t) \otimes h_{LP}(t) + W(t) \otimes h_{LP}(t)$$

- As concerning the useful signal:

$$X(t) \otimes h_{LP}(t) = S(t) \otimes h(t) \otimes h_{LP}(t) \cong S(t) \otimes h(t) = X(t)$$

- As concerning the noise component:

$$W_B(t) = W(t) \otimes h_{LP}(t) \quad \Rightarrow \quad S_{W_B}(f) = S_W(f) \left| H_{LP}(f) \right|^2 = \frac{N_0}{2} rect\left(\frac{f}{2B} \right)$$

Example: Linear deconvolution

- We sample at intervals T_c=1/(2B) and we collect N=2BT samples:

$$Z_k \triangleq \frac{1}{\sqrt{2B}}\left[Z(t)\otimes h_{LP}(t)\right]\Big|_{t=\frac{k}{2B}} = \frac{1}{\sqrt{2B}}X\left(\frac{k}{2B}\right)+\frac{1}{\sqrt{2B}}W_B\left(\frac{k}{2B}\right)$$

$$\left\{W_k \triangleq \frac{1}{\sqrt{2B}}W_B\left(\frac{k}{2B}\right)\right\}_{k=0}^{N-1} \quad \text{uncorrelated, Gaussian distributed, zero mean}$$

$$\text{and variance } \sigma_W^2=\frac{N_0}{2} \quad \rightarrow \quad W_k \in \mathcal{N}\left(0,\frac{N_0}{2}\right), \quad \text{IID}$$

$$X_k \triangleq \frac{1}{\sqrt{2B}}X\left(\frac{k}{2B}\right)=\frac{1}{\sqrt{2B}}\left[\int_0^T S(\tau)h(t-\tau)d\tau\right]\Bigg|_{t=\frac{k}{2B}}, \quad k=0,1,2,\cdots,N-1$$

Example: Linear deconvolution

$$X_k \triangleq \frac{1}{\sqrt{2B}} X\left(\frac{k}{2B}\right) = \frac{1}{\sqrt{2B}} \int_0^T S(\tau) h\left(\frac{k}{2B} - \tau\right) d\tau$$

- Approximating the integral with the rule of rectangles:

$$X_k \cong \frac{1}{\sqrt{2B}} \sum_{i=0}^{N_p - 1} S\left(i\Delta T\right) h\left(\frac{k}{2B} - i\Delta T\right) \cdot \Delta T, \quad \text{where } N_p = \frac{T}{\Delta T}$$

- The approximation is accurate if $N_p >> 1$. Let us assume $\Delta T = 1/2B$, so we get $N_p = N = 2BT$:

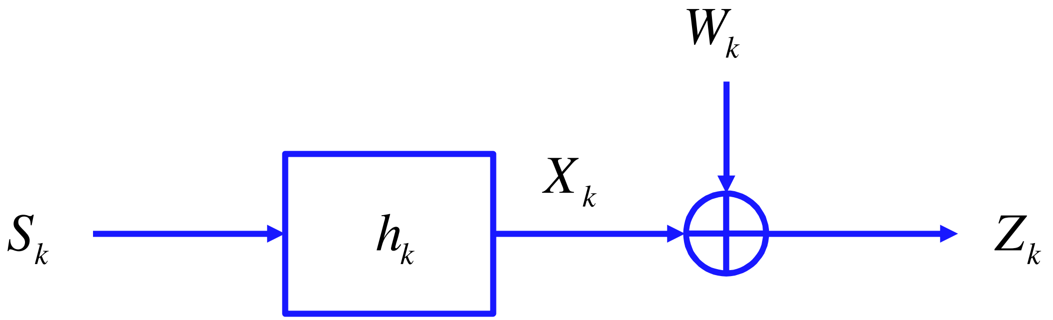
$$X_k \cong \sum_{i=0}^{N-1} \frac{1}{\sqrt{2B}} S\left(\frac{i}{2B}\right) h\left(\frac{k}{2B} - \frac{i}{2B}\right) \cdot \frac{1}{2B}, \quad \text{where } N = 2BT$$

- Now, define: $S_k \triangleq \frac{1}{\sqrt{2B}} S\left(\frac{k}{2B}\right), \qquad h_k \triangleq \frac{1}{2B} h\left(\frac{k}{2B}\right)$

Example: Linear deconvolution

- we get the **equivalent discrete-time system**:

$$X_k \cong \sum_{i=0}^{N-1} S_i h_{k-i} = S_k \otimes h_k \quad \Rightarrow \quad Z_k = X_k + W_k = S_k \otimes h_k + W_k$$



- If $S(t)$ has duration $T_s<T$, then S_k has duration $N_s=T_s/T_c=2BT_s<N$
- If $h(t)$ has duration $T_h<T$, then h_k has duration $N_h=T_h/T_c=2BT_h<N$
- The convolution between S_k and h_k has duration N_s+N_h-1 so $N \geq N_s+N_h-1$

Example: Linear deconvolution

- Let us assume that $T = T_s + T_h$, so that $N = N_s + N_h - 1$
- Recalling that $S_i = 0$ if $i < 0$ or $i \geq N_s$, the observed data samples are given by:

$$
\begin{aligned}
Z_k &= \sum_{i=0}^{N_s-1} S_i h_{k-i} + W_k, \quad k = 0,1,2,\cdots,N-1 \\
&= S_0 h_k + S_1 h_{k-1} + \cdots + S_{N_s-1} h_{k-(N_s-1)} + W_k
\end{aligned}
$$

$$
\begin{cases}
k = 0: & Z_0 = S_0 h_0 + S_1 h_{-1} + \cdots + S_{N_s-1} h_{-(N_s-1)} + W_0 \\
k = 1: & Z_1 = S_0 h_1 + S_1 h_0 + \cdots + S_{N_s-1} h_{1-(N_s-1)} + W_1 \\
\quad\vdots & \\
k = N-1: & Z_{N-1} = S_0 h_{N-1} + S_1 h_{N-2} + \cdots + S_{N_s-1} h_{N-(N_s-1)} + W_{N-1}
\end{cases}
$$

Example: Linear deconvolution

- In vector format (keeping in mind that h_k is a causal filter with memory N_h, i.e. $h_k = 0$ for $k<0$ and $k \geq N_h$):

$N = N_s + N_h - 1$

$$\underbrace{\begin{bmatrix} Z_0 \\ Z_1 \\ \vdots \\ Z_{N-1} \end{bmatrix}}_{\mathbf{z}} = \underbrace{\begin{bmatrix} h_0 & 0 & \cdots & \cdots & 0 & 0 \\ h_1 & h_0 & \ddots & \ddots & \vdots & 0 \\ h_2 & h_1 & \ddots & \ddots & 0 & \vdots \\ \vdots & h_2 & \ddots & \ddots & h_0 & 0 \\ h_{N_h-1} & \vdots & \ddots & \ddots & h_1 & h_0 \\ 0 & h_{N_h-1} & \ddots & \ddots & \vdots & h_1 \\ \vdots & \ddots & \cdots & \cdots & h_{N_h-1} & \vdots \\ 0 & 0 & \cdots & \cdots & 0 & h_{N_h-1} \end{bmatrix}}_{\underset{N \times N_s}{\mathbf{H}}} \underbrace{\begin{bmatrix} S_0 \\ S_1 \\ \vdots \\ S_{N_s-1} \end{bmatrix}}_{\mathbf{s}} + \underbrace{\begin{bmatrix} W_0 \\ W_1 \\ \vdots \\ W_{N-1} \end{bmatrix}}_{\mathbf{w}}$$

$$\underset{N\times 1}{\mathbf{Z}} = \underset{N\times N_s}{\mathbf{H}} \cdot \underset{N_s\times 1}{\mathbf{S}} + \underset{N\times 1}{\mathbf{W}}, \quad \text{where} \quad N = N_s + N_h - 1 > N_s$$

Example: Linear deconvolution

- Hence, the problem is exactly in the form of a **linear signal model** that we have already investigated, both the case of deterministic signal (**S** is a deterministic unknown vector) and the case of random signal (**S** is a random vector).

- Assume now that $S(t)$ is a WSS random process, not necessarily Gaussian, with zero mean function and known autocorrelation function (ACF) $R_s(\tau)$.

- In this case, we get the **Bayesian linear model**: **Z=HS+W**

$$\mathbf{W} \in \mathcal{N}\left(\mathbf{0}, \frac{N_0}{2}\mathbf{I}\right), \quad \mathbf{H} \text{ known}, \quad \boldsymbol{\eta}_s = E\{\mathbf{S}\} = \mathbf{0}, \quad \mathbf{C}_s = E\{\mathbf{S}\mathbf{S}^T\}$$

$$[\mathbf{C}_s]_{i+1,k+1} = E\{S_i S_k\} = E\left\{\frac{1}{\sqrt{2B}} S\left(\frac{i}{2B}\right)\frac{1}{\sqrt{2B}} S\left(\frac{k}{2B}\right)\right\} = \frac{1}{2B} R_s\left(\frac{k-i}{2B}\right)$$

$$i,k = 0,1,2,\cdots,N_s - 1$$

Example: Linear deconvolution

- The LMMSE estimator of **S** given **Z** is known:

$$
\begin{cases}
\hat{\mathbf{S}}_{LMMSE} = \mathbf{C}_s \mathbf{H}^T \left(\mathbf{H} \mathbf{C}_s \mathbf{H}^T + \dfrac{N_0}{2} \mathbf{I} \right)^{-1} \mathbf{Z} \\
\\
\mathbf{C}_\varepsilon = \left(\mathbf{C}_s^{-1} + \dfrac{2}{N_0} \mathbf{H}^T \mathbf{H} \right)^{-1} \quad \rightarrow \quad MSE\left\{ \hat{\mathbf{S}}_{i,LMMSE} \right\} = \left[\mathbf{C}_\varepsilon \right]_{i,i}
\end{cases}
$$

- If $S(t)$ is a Gaussian process, then **S** is a Gaussian vector and the LMMSE estimator is also the MMSE (and MAP) estimator.

Example: Linear deconvolution

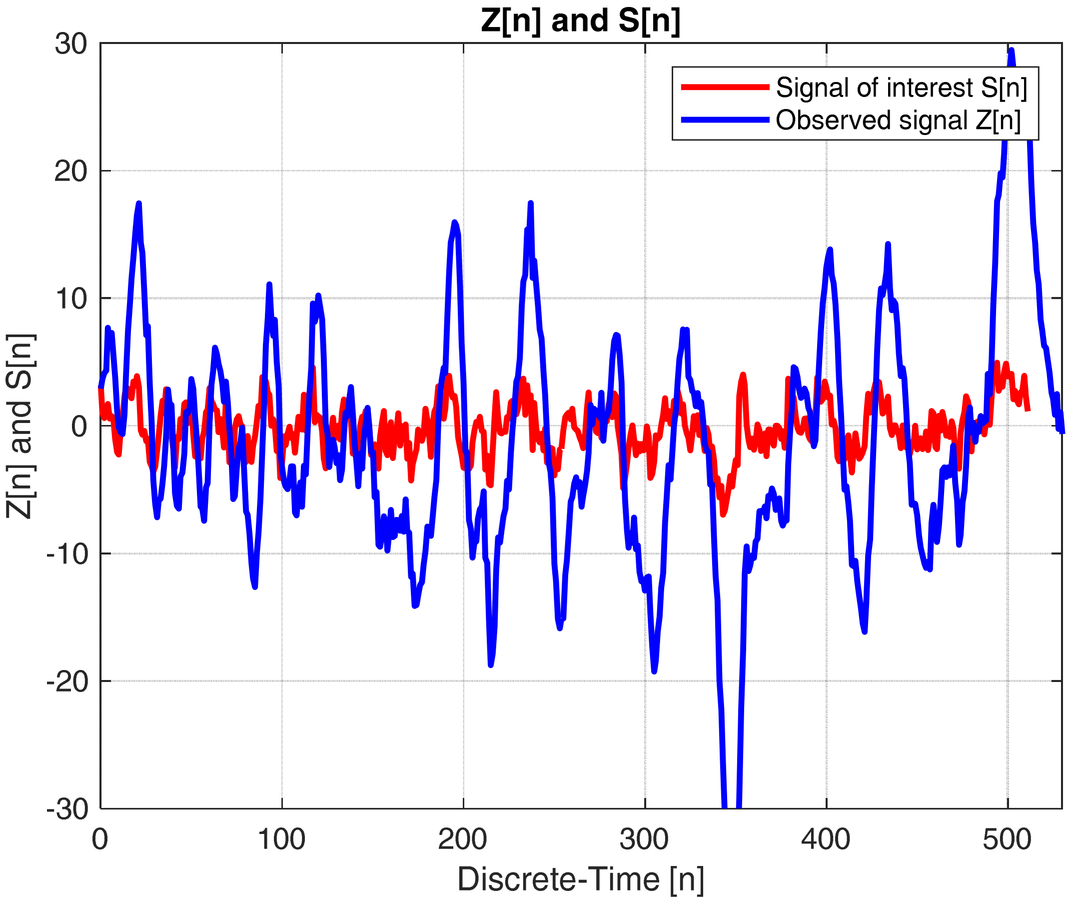
- **Numerical Example**:
- S_k is a discrete-time low-pass AR(1) Gaussian process with one-lag correlation coefficient ρ=0.8 and power $\sigma_s^2 = 5$
- h_k is a FIR(N_h-1) filter with impulse response $h[n]=\alpha^n$ for $0 \le n \le N_h-1$, with α=0.9 and N_h=20.
- W_k is AWGN with power $\sigma_W^2 = \dfrac{N_0}{2}$

$$R_s[k-i] = \sigma_S^2 \rho^{|k-i|}, \quad S_s\left(e^{j2\pi f}\right) = \frac{\sigma_s^2 (1-\rho^2)}{1+\rho^2 - 2\rho\cos\left(2\pi f\right)}$$

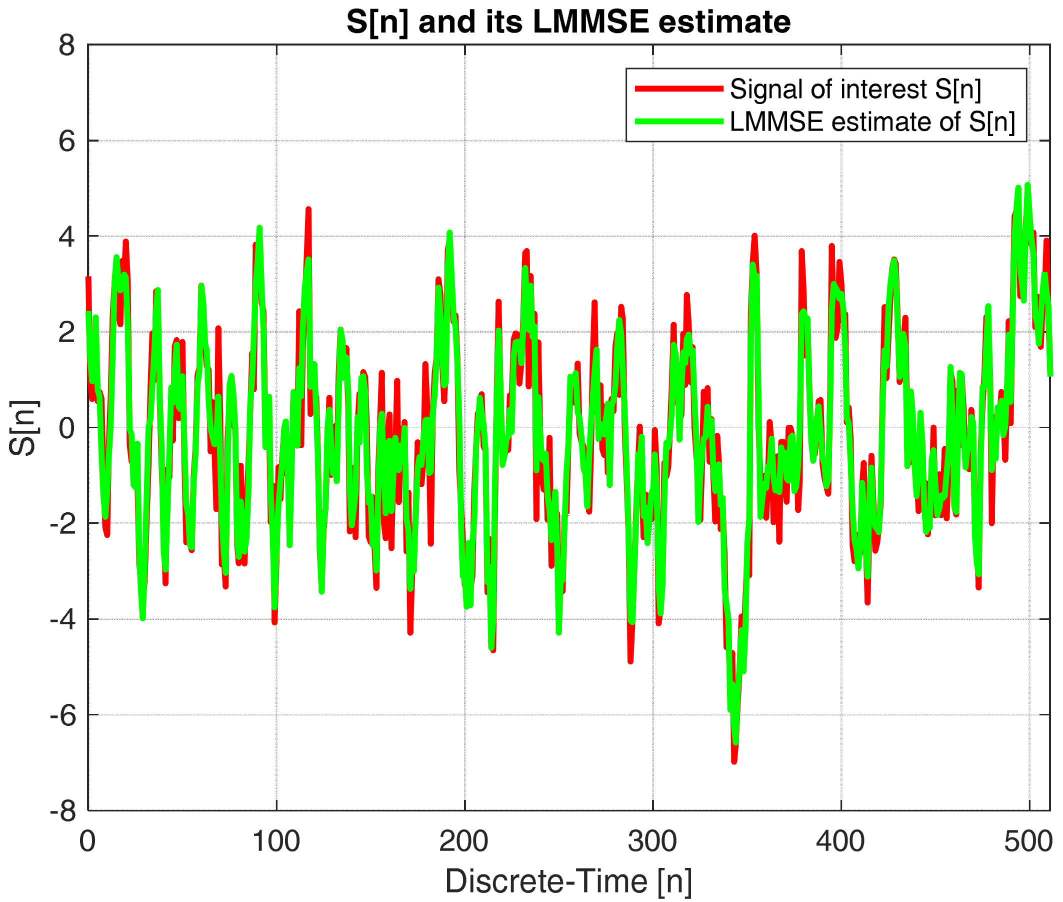
$$\left[\mathbf{C}_s\right]_{i+1,k+1} = E\left\{S_i S_k\right\} = R_s[k-i] = \sigma_S^2 \rho^{|k-i|}, \quad i,k = 0,1,2,\cdots,N_s-1$$

$$N_S = 512, \quad N = N_s + N_h - 1 = 531, \quad \gamma_{dB} = 10 Log_{10}\left(\frac{\sigma_s^2}{\sigma_W^2}\right) = 10\ dB$$

Example: Linear deconvolution



Example: Linear deconvolution



Example: Linear deconvolution

- If the channel is *transparent*, $h_k=\delta_k$ (and so $N=N_s$), then **H**=**I**, and so we get:

$$\hat{\mathbf{S}}_{LMMSE} = \mathbf{C}_s\left(\mathbf{C}_s + \frac{N_0}{2}\mathbf{I}\right)^{-1}\mathbf{Z} = \mathbf{A}\cdot\mathbf{Z}$$

$$\underset{N_s \times N_s}{\mathbf{A}} \triangleq \mathbf{C}_s\left(\mathbf{C}_s + \frac{N_0}{2}\mathbf{I}\right)^{-1} \quad \text{Wiener Smoothing Filter (batch)}$$

- Scalar case: $N_s=1$ $\quad \hat{S}_{0,LMMSE} = \alpha Z_0 = \frac{\gamma}{\gamma+1}Z_0$

$$\alpha \triangleq \sigma_s^2\left(\sigma_s^2 + \frac{N_0}{2}\right)^{-1} = \frac{\sigma_s^2}{\sigma_s^2 + \frac{N_0}{2}} = \frac{\gamma}{\gamma+1}, \quad \text{where} \quad \gamma \triangleq \frac{\sigma_s^2}{N_0/2} \quad \text{is the SNR}$$

Example: Linear deconvolution

- Scalar case: N_s=1 $\qquad \hat{S}_{0,LMMSE} = \dfrac{\gamma}{\gamma+1} Z_0$

$$\gamma \gg 1 \qquad \Rightarrow \qquad \hat{S}_{0,LMMSE} \cong Z_0$$

$$\gamma \ll 1 \qquad \Rightarrow \qquad \hat{S}_{0,LMMSE} \cong 0 \quad (= \text{mean value of } S_0)$$

- **Remark:** we talk about **blind deconvolution** when also the matrix **H** is unknown, i.e. we do not know the channel impulse response $h(t)$.
- **Remark**: we talk about **channel identification** when **S** is known and the problem is to estimate $h(t)$, i.e. **H**; we talk about **blind channel identification** when we want to estimate **H** and also **S** is unknown (in this case typically we need to know a priori some statistics of the signal **S**, e.g. the ACF).

Example: Channel identification

- **Channel identification:** $\{S_i\}$ is known (S_i=0 for i<0 or $i \geq N_s$) and the problem is to estimate $\{h_i\}$.

- Recalling that the filter is causal and with finite memory, i.e. h_i=0 for i<0 or $i \geq N_h$, the observed data samples are given by:

$$Z_k = \sum_{i=0}^{N_s-1} S_i h_{k-i} + W_k = \sum_{i=0}^{N_h-1} h_i S_{k-i} + W_k$$

$$= h_0 S_k + h_1 S_{k-1} + \cdots + h_{N_h-1} S_{k-(N_h-1)} + W_k, \quad k = 0,1,2,\cdots,N-1$$

$$\begin{cases} k=0: \quad Z_0 = h_0 S_0 + h_1 S_{-1} + \cdots + h_{N_h-1} S_{-(N_h-1)} + W_0 \\ k=1: \quad Z_1 = h_0 S_1 + h_1 S_0 + \cdots + h_{N_h-1} S_{1-(N_h-1)} + W_1 \\ \quad \vdots \\ k=N-1: \quad Z_{N-1} = h_0 S_{N-1} + h_1 S_{N-2} + \cdots + h_{N_h-1} S_{N-(N_h-1)} + W_{N-1} \end{cases}$$

$N = N_s + N_h - 1$

Example: Channel identification

$N = N_s + N_h - 1$

$$\underbrace{\begin{bmatrix} z_0 \\ z_1 \\ \vdots \\ z_{N-1} \end{bmatrix}}_{\mathbf{z}} = \underbrace{\begin{bmatrix} s_0 & 0 & \cdots & \cdots & 0 & 0 \\ s_1 & s_0 & \ddots & \ddots & \vdots & 0 \\ s_2 & s_1 & \ddots & \ddots & 0 & \vdots \\ \vdots & s_2 & \ddots & \ddots & s_0 & 0 \\ s_{N_s-1} & \vdots & \ddots & \ddots & s_1 & s_0 \\ 0 & s_{N_s-1} & \ddots & \ddots & \vdots & s_1 \\ \vdots & \ddots & \cdots & \cdots & s_{N_s-1} & \vdots \\ 0 & 0 & \cdots & \cdots & 0 & s_{N_s-1} \end{bmatrix}}_{\substack{\mathbf{S} \\ N \times N_h}} \underbrace{\begin{bmatrix} h_0 \\ h_1 \\ \vdots \\ h_{N_h-1} \end{bmatrix}}_{\mathbf{h}} + \underbrace{\begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_{N-1} \end{bmatrix}}_{\mathbf{w}}$$

$$\underset{N\times 1}{\mathbf{z}} = \underset{N\times N_h}{\mathbf{S}} \cdot \underset{N_h\times 1}{\mathbf{h}} + \underset{N\times 1}{\mathbf{w}}, \quad \text{where} \quad N = N_s + N_h - 1 > N_h$$

Example: Channel identification

$$\underset{N\times 1}{\mathbf{Z}} = \underset{N\times N_h}{\mathbf{S}} \cdot \underset{N_h\times 1}{\mathbf{h}} + \underset{N\times 1}{\mathbf{W}}, \quad \text{where} \quad N = N_s + N_h - 1 > N_h$$

- The **LMMSE estimator** of **h** given **Z** is given by:

$$\hat{\mathbf{h}}_{LMMSE} = \mathbf{C}_h \mathbf{S}^T \left(\mathbf{S}\mathbf{C}_h \mathbf{S}^T + \frac{N_0}{2}\mathbf{I} \right)^{-1} \mathbf{Z}$$

- If the channel is deterministic (so we do not have $\mathbf{C}_h$), we can derive the **linear LS estimator** of **h**:

$$\hat{\mathbf{h}}_{LS} = \mathbf{S}^{\#}\mathbf{Z} = (\mathbf{S}^T\mathbf{S})^{-1}\mathbf{S}^T\mathbf{Z}$$

Sequential (Recursive) LMMSE Estimator

- The **LMMSE** estimator can also be expressed in **sequential** (non batch) form, where the estimate is updated each time a new sample is acquired, without re-doing the same processing also on the previously acquired data.

- Let us analyze this for the case where we want to estimate a constant random signal *S* embedded in additive white noise *W*[*n*].

$$X[n] = S + W[n], \quad n = 0, 1, 2, \cdots, N-1$$

$$\boldsymbol{\theta} = S, \quad \boldsymbol{\eta}_\theta = \eta_S, \quad \mathbf{C}_\theta = \sigma_S^2, \quad \mathbf{H} = \mathbf{1}, \quad \mathbf{C}_W = \sigma_W^2 \mathbf{I}$$

$$\begin{cases} \hat{S}_{LMMSE} = \boldsymbol{\eta}_\theta + \mathbf{C}_\theta \mathbf{H}^T \left(\mathbf{H}\mathbf{C}_\theta \mathbf{H}^T + \mathbf{C}_W \right)^{-1} \left(\mathbf{X} - \mathbf{H}\boldsymbol{\eta}_\theta \right) \\ MSE\left\{ \hat{S}_{LMMSE} \right\} = \left(\mathbf{C}_\theta^{-1} + \mathbf{H}^T \mathbf{C}_W^{-1} \mathbf{H} \right)^{-1} \end{cases}$$

Sequential (Recursive) LMMSE Estimator

- The **LMMSE** estimator of S in "batch" mode and its MSE are given by:

$$\hat{S}_{LMMSE} = \alpha \cdot \frac{1}{N}\mathbf{1}^T\mathbf{X} + (1-\alpha)\eta_S = \alpha\bar{\mathbf{X}} + (1-\alpha)\eta_S = \eta_S + \alpha(\bar{\mathbf{X}} - \eta_S)$$

$$\bar{\mathbf{X}} \triangleq \frac{1}{N}\mathbf{1}^T\mathbf{X} = \frac{1}{N}\sum_{n=0}^{N-1} X[n]$$

$$MSE\left\{\hat{S}_{LMMSE}\right\} = \left(\mathbf{C}_\theta^{-1} + \mathbf{H}^T\mathbf{C}_W^{-1}\mathbf{H}\right)^{-1} = \frac{N\sigma_S^2}{\sigma_W^2 + N\sigma_S^2} \cdot \frac{\sigma_W^2}{N} = \alpha \cdot \frac{\sigma_W^2}{N}$$

$$\text{where} \quad \alpha \triangleq \frac{N\sigma_S^2}{\sigma_W^2 + N\sigma_S^2} = \frac{N\gamma}{1+N\gamma}, \quad \gamma \triangleq \frac{\sigma_S^2}{\sigma_W^2}$$

Sequential (Recursive) LMMSE Estimator

- Let us express the **LMMSE** estimator of *S* putting in evidence that the estimate has been done at the discrete-time *N*-1, i.e. by making use of the *N* data samples X[0], X[1], … , X[*N*-1].

$$\begin{cases} \hat{S}[N-1] = \dfrac{N\gamma}{1+N\gamma}\bar{\mathbf{X}}[N-1] + \dfrac{1}{1+N\gamma}\eta_S \\ MSE[N-1] = \dfrac{\gamma\sigma_W^2}{1+N\gamma} \end{cases}$$

where

$$\bar{\mathbf{X}}[N-1] \triangleq \frac{1}{N}\sum_{n=0}^{N-1} X[n]$$

$$\gamma \triangleq \frac{\sigma_S^2}{\sigma_W^2}$$

Sequential (Recursive) LMMSE Estimator

- At discrete-time N we acquire a new data sample $\rightarrow$ $X[N]$

$$
\begin{cases}
\hat{S}[N] = \dfrac{(N+1)\gamma}{1+(N+1)\gamma}\bar{\mathbf{X}}[N] + \dfrac{1}{1+(N+1)\gamma}\eta_S \\
MSE[N] \triangleq \dfrac{\gamma\sigma_W^2}{1+(N+1)\gamma}
\end{cases}
$$

where

$$
\bar{\mathbf{X}}[N] = \frac{1}{N+1}\sum_{n=0}^{N} X[n], \quad \gamma \triangleq \frac{\sigma_S^2}{\sigma_W^2}
$$

- Can we express the estimate and the MSE at discrete-time N as a function of the estimate and MSE at discrete-time N-1 and of the new data sample $X[N]$?

Sequential (Recursive) LMMSE Estimator

- Let's first express the estimate at step *N* as a function of the estimate (already calculated) at step *N*-1:

$$\bar{\mathbf{X}}[N]=\frac{1}{N+1}\sum_{n=0}^{N}X[n]=\frac{1}{N+1}X[N]+\frac{1}{N+1}\sum_{n=0}^{N-1}X[n]$$

$$=\frac{1}{N+1}X[N]+\frac{N}{N+1}\cdot\frac{1}{N}\sum_{n=0}^{N-1}X[n]$$

$$=\frac{1}{N+1}X[N]+\frac{N}{N+1}\bar{\mathbf{X}}[N-1]$$

$$\hat{S}[N]=\frac{(N+1)\gamma}{1+(N+1)\gamma}\overbrace{\left(\frac{1}{N+1}X[N]+\frac{N}{N+1}\bar{\mathbf{X}}[N-1]\right)}^{\bar{\mathbf{X}}[N]}+\frac{1}{1+(N+1)\gamma}\eta_S$$

$$=\frac{\gamma}{1+(N+1)\gamma}X[N]+\frac{N\gamma}{1+(N+1)\gamma}\bar{\mathbf{X}}[N-1]+\frac{1}{1+(N+1)\gamma}\eta_S$$

$$=\frac{\gamma}{1+N\gamma+\gamma}X[N]+\frac{1+N\gamma}{1+N\gamma+\gamma}\cdot\frac{N\gamma}{1+N\gamma}\bar{\mathbf{X}}[N-1]+\frac{1+N\gamma}{1+N\gamma+\gamma}\cdot\frac{1}{1+N\gamma}\eta_S$$

Sequential (Recursive) LMMSE Estimator

$$\hat{S}[N] = \frac{\gamma}{1+N\gamma+\gamma}X[N] + \frac{1+N\gamma}{1+N\gamma+\gamma}\cdot\left(\frac{N\gamma}{1+N\gamma}\bar{\mathbf{X}}[N-1] + \frac{1}{1+N\gamma}\eta_S\right)$$

$$= \frac{\gamma}{1+N\gamma+\gamma}X[N] + \frac{1+N\gamma}{1+N\gamma+\gamma}\hat{S}[N-1]$$

$$= \frac{\gamma}{1+N\gamma+\gamma}X[N] + \left(1 - \frac{\gamma}{1+N\gamma+\gamma}\right)\hat{S}[N-1]$$

$$= \hat{S}[N-1] + \frac{\gamma}{1+N\gamma+\gamma}\left(X[N] - \hat{S}[N-1]\right)$$

$I[N] \triangleq X[N] - \hat{S}[N-1] \quad \rightarrow \quad$ Innovation at step N

$K[N] \triangleq \dfrac{\gamma}{1+N\gamma+\gamma} \quad \rightarrow \quad$ *Gain factor* at step N

Sequential (Recursive) LMMSE Estimator

- It can be proved that the random variables $\{I[n]\}$ are uncorrelated, i.e. $I[n]$ is a discrete-time white process, which represents the new information (**innovation**) contained in the data sample $X[n]$ with respect to the past.

$$MSE[N-1] = \frac{\gamma\sigma_W^2}{1+N\gamma}$$

- *Gain factor* at step N vs. MSE[N-1]:

$$K[N] = \frac{\gamma}{1+N\gamma+\gamma} = \frac{\gamma\sigma_W^2}{1+N\gamma} \cdot \frac{1+N\gamma}{(1+N\gamma+\gamma)\sigma_W^2}$$

$$= MSE[N-1] \cdot \frac{1+N\gamma}{(1+N\gamma)\sigma_W^2 + \gamma\sigma_W^2}$$

$$= MSE[N-1] \cdot \frac{1}{\sigma_W^2 + \dfrac{\gamma\sigma_W^2}{1+N\gamma}} = \frac{MSE[N-1]}{\sigma_W^2 + MSE[N-1]}$$

Sequential (Recursive) LMMSE Estimator

$$\textit{Gain factor} \text{ at step } N \textit{ vs } MSE[N-1]:$$

$$K[N] = \frac{\gamma}{1 + N\gamma + \gamma} = \frac{MSE[N-1]}{MSE[N-1] + \sigma_W^2}$$

- $K[N]$ tends to 0 when $N \to \infty$.
- This is because when the data size N increases the estimate obtained by processing previous data tends to be more and more accurate (the estimator is consistent) and so the new data sample tends to be less and less useful. Finally, if the signal is constant, the estimate converges in probability to the true value.

Sequential (Recursive) LMMSE Estimator

- Let us know express the MSE at step *N* as a function of the MSE already calculated at step *N*-1:

$$MSE[N] = \frac{\gamma\sigma_W^2}{1+(N+1)\gamma} = \frac{1+N\gamma}{1+N\gamma+\gamma} \cdot \frac{\gamma\sigma_W^2}{1+N\gamma}$$

$$= \frac{1+N\gamma}{1+N\gamma+\gamma} MSE[N-1] = \frac{1+N\gamma+\gamma-\gamma}{1+N\gamma+\gamma} MSE[N-1]$$

$$= \left(1 - \frac{\gamma}{1+N\gamma+\gamma}\right) MSE[N-1]$$

$$= \left(1 - K[N]\right) MSE[N-1]$$

Sequential (Recursive) LMMSE Estimator

- In summary, the **LMMSE** estimate of *S* can be calculated **recursively** in two steps:

Initialization:

$$\hat{S}[-1] = \eta_S \qquad MSE[-1] = \sigma_S^2$$

Estimator Update:

$$\begin{cases} K[N] = \dfrac{MSE[N-1]}{MSE[N-1] + \sigma_W^2} \\ \\ \hat{S}[N] = \hat{S}[N-1] + K[N]\left(X[N] - \hat{S}[N-1]\right) \end{cases}$$

Minimum MSE Update:

$$MSE[N] = \left(1 - K[N]\right) MSE[N-1]$$

Sequential (Recursive) LMMSE Estimator

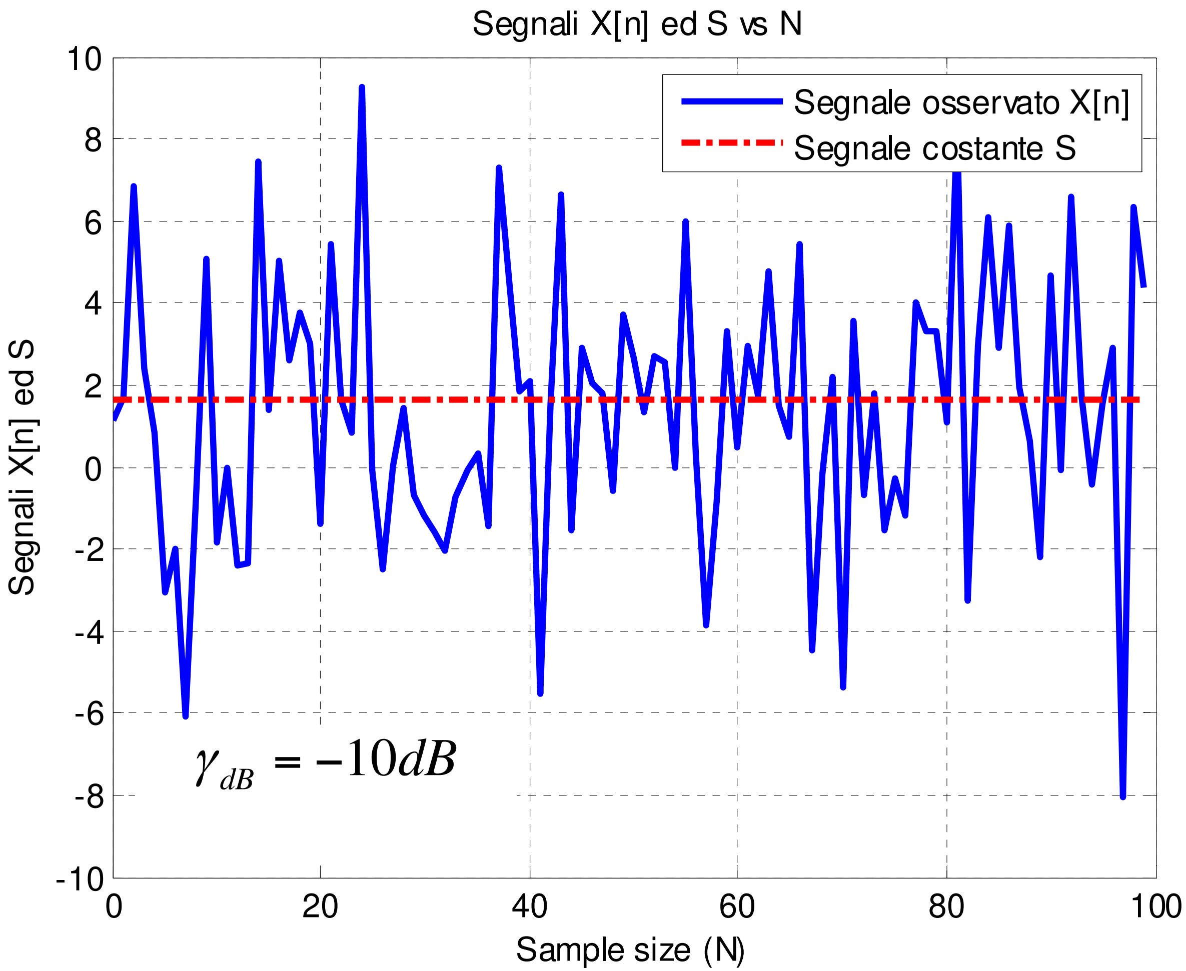
- In a similar manner we could derive the recursive LMMSE estimator of the random vector θ given the observed data $X[0]$, $X[1]$, … , $X[N\text{-}1]$, …. with increasing N:

- This recursive implementation of the LMMSE estimator is the idea that is at the base of the **Kalman filter**, which is the recursive LMMSE estimator of the random signal $S[n]$ observed in additive Gaussian noise, when $S[n]$ is a Markov process, i.e. an AR(1) process.

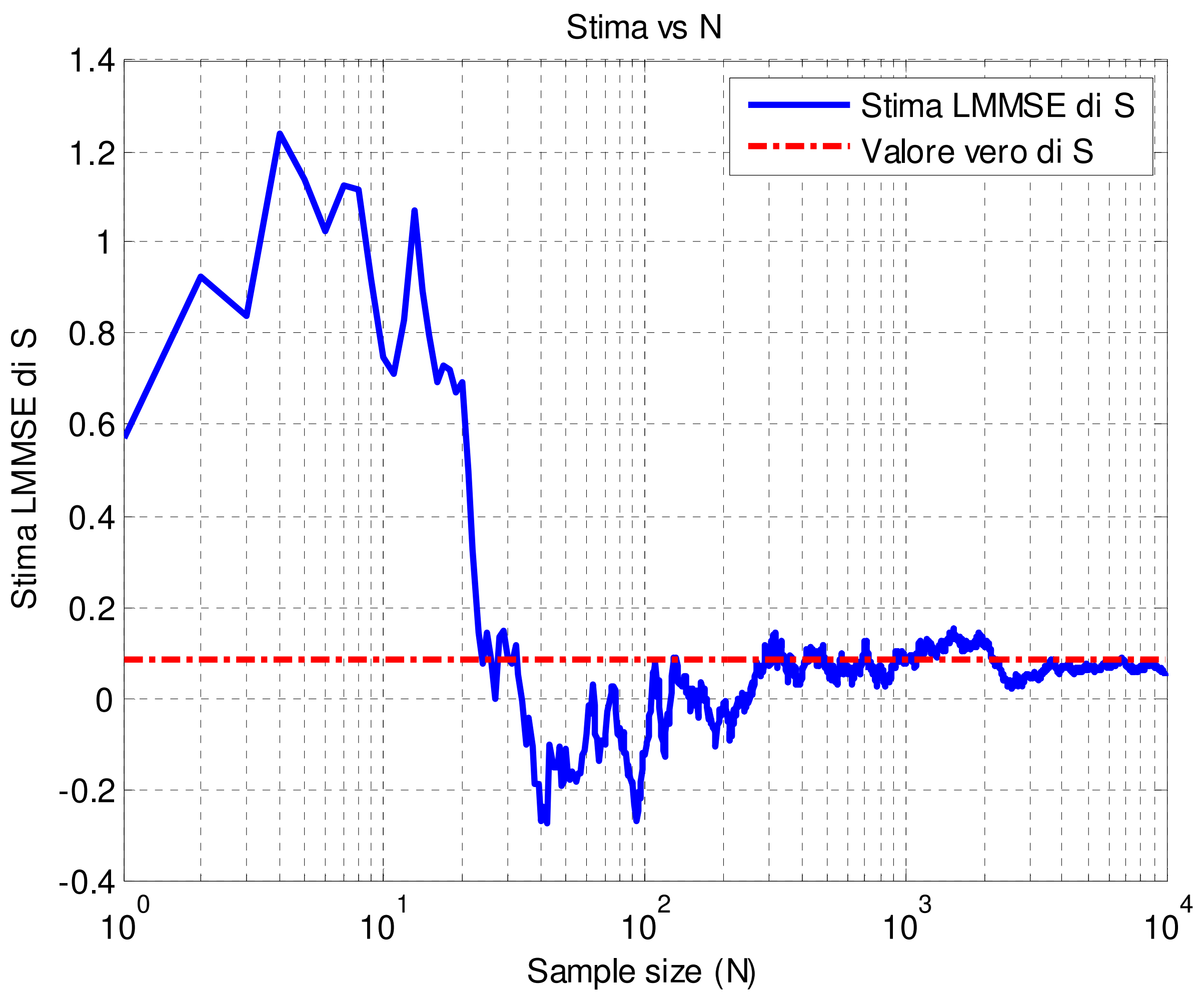
Numerical example:

$$\begin{cases} \eta_S = 1 \\ \sigma_S^2 = 1 \\ N_{\max} = 10^4 \\ \gamma_{dB} = 10 Log_{10}\left(\sigma_S^2 / \sigma_W^2\right) \end{cases}$$

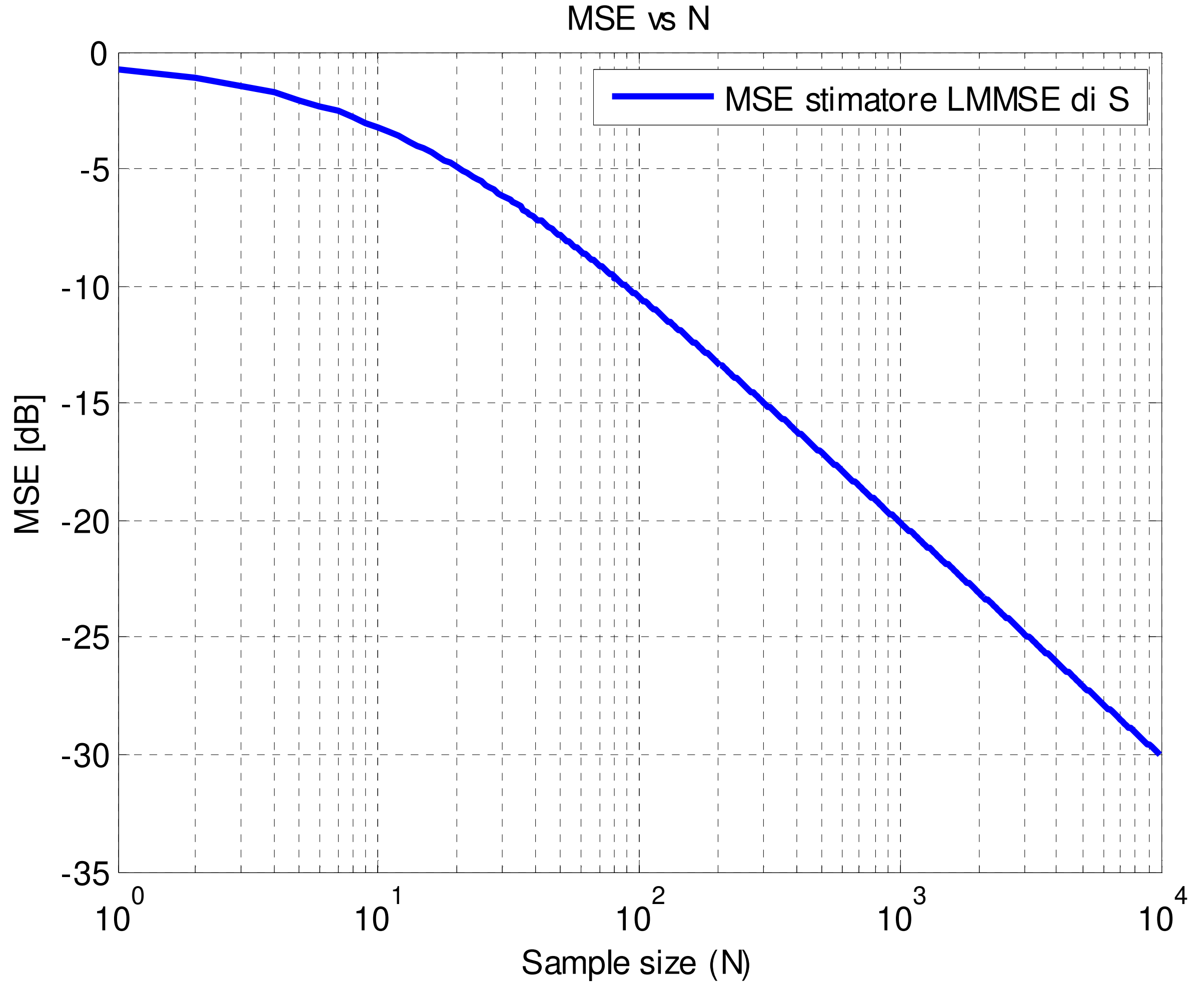
Sequential (Recursive) LMMSE Estimator



Sequential (Recursive) LMMSE Estimator

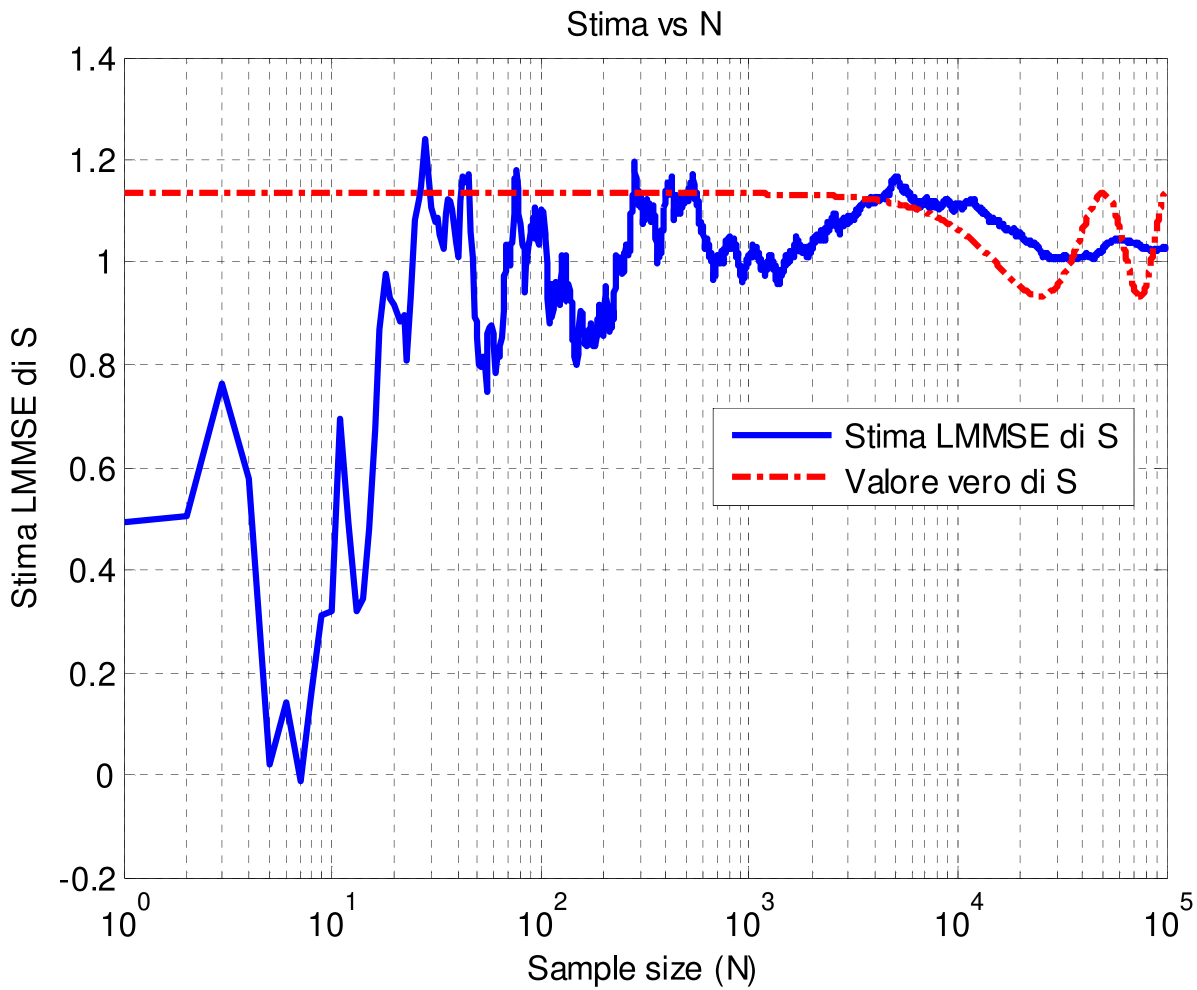


Sequential (Recursive) LMMSE Estimator



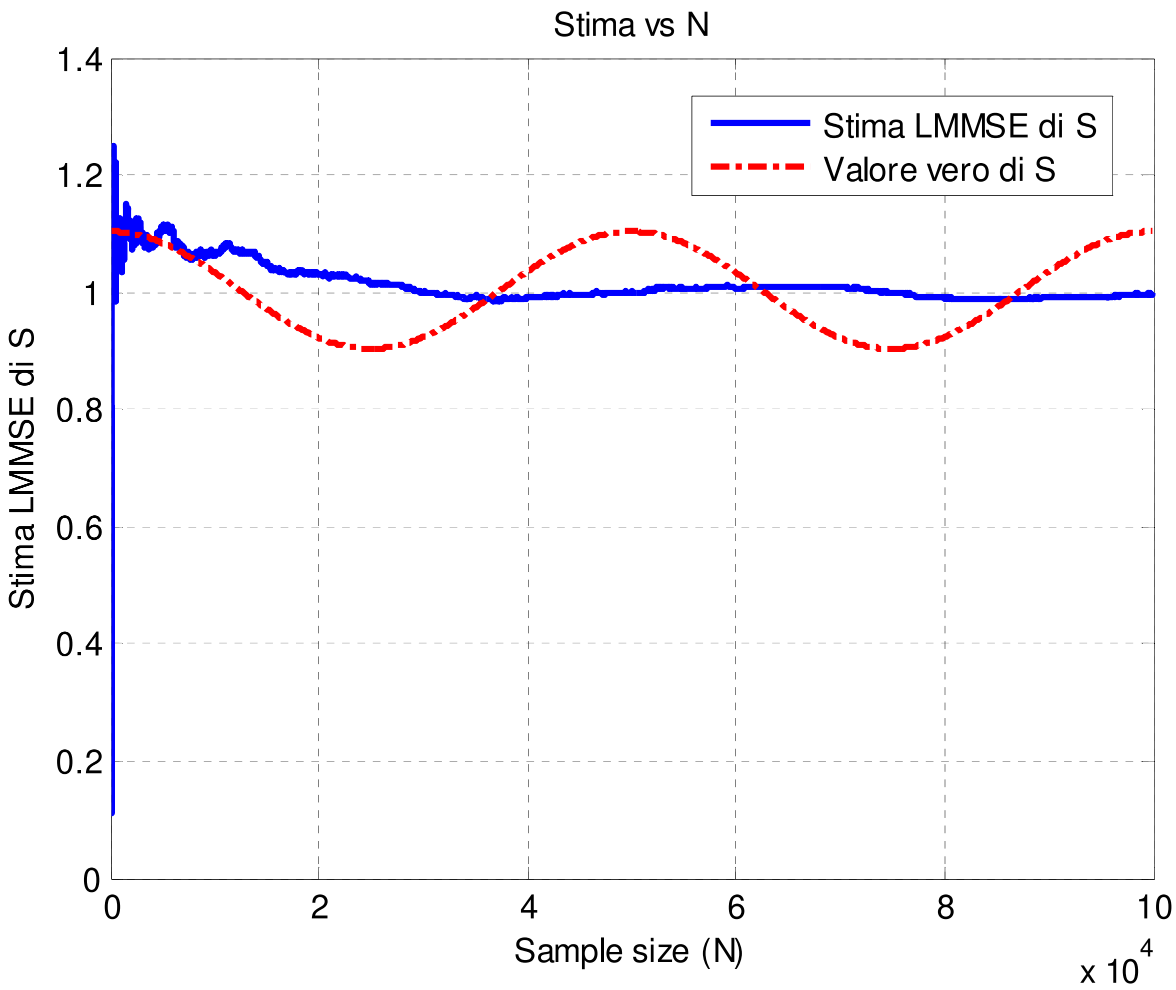
Sequential (Recursive) LMMSE Estimator

- If the signal of interest (SOI) is not constant, the **LMMSE** estimator we derived (which is no more the optimal one) tracks the variations, since it basically estimates the local mean of the observed signal (low-pass effect → it has some inertia).



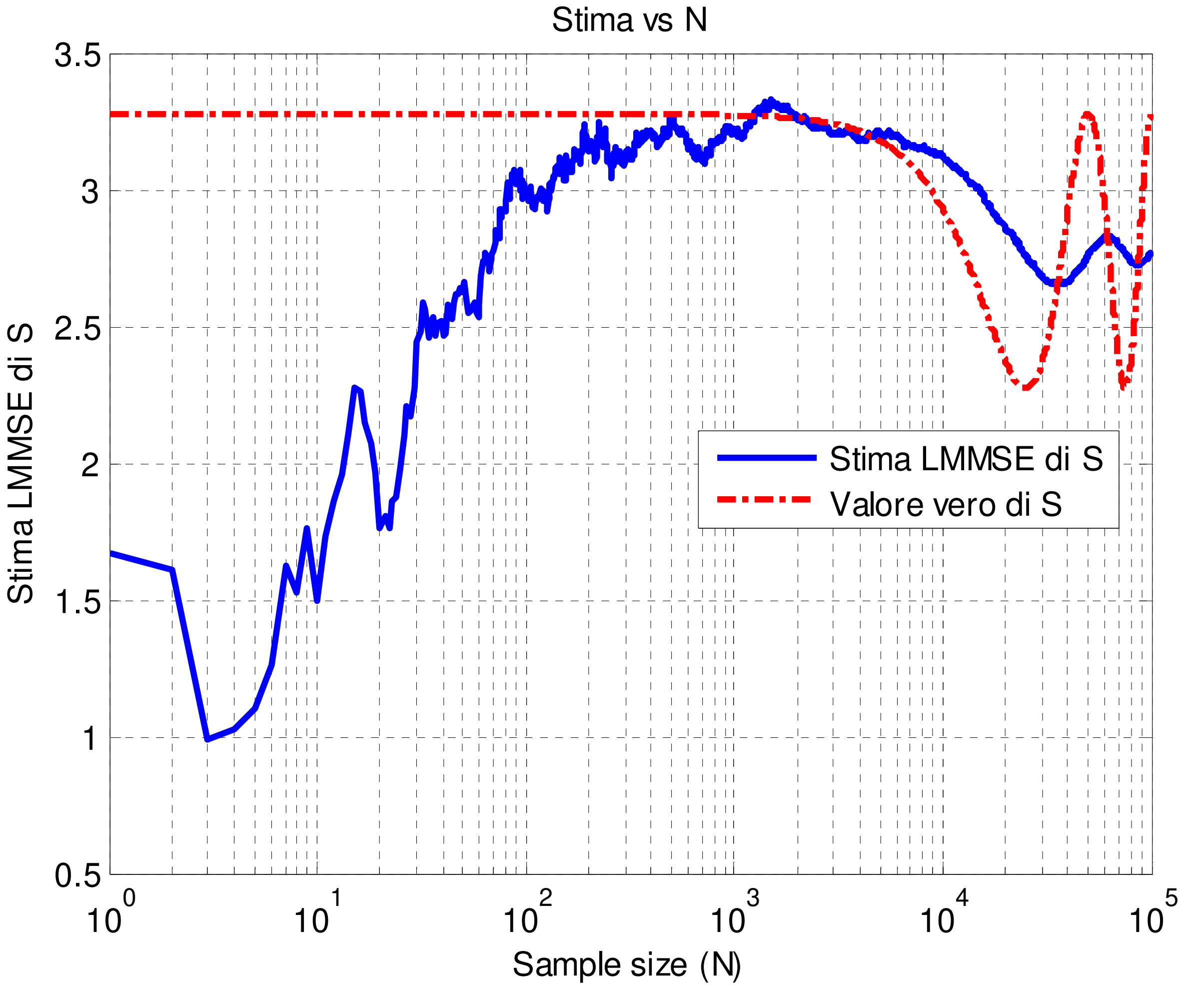
Sequential (Recursive) LMMSE Estimator

- The low-pass effect is more evident if we plot the estimate with the discrete-time in linear scale:



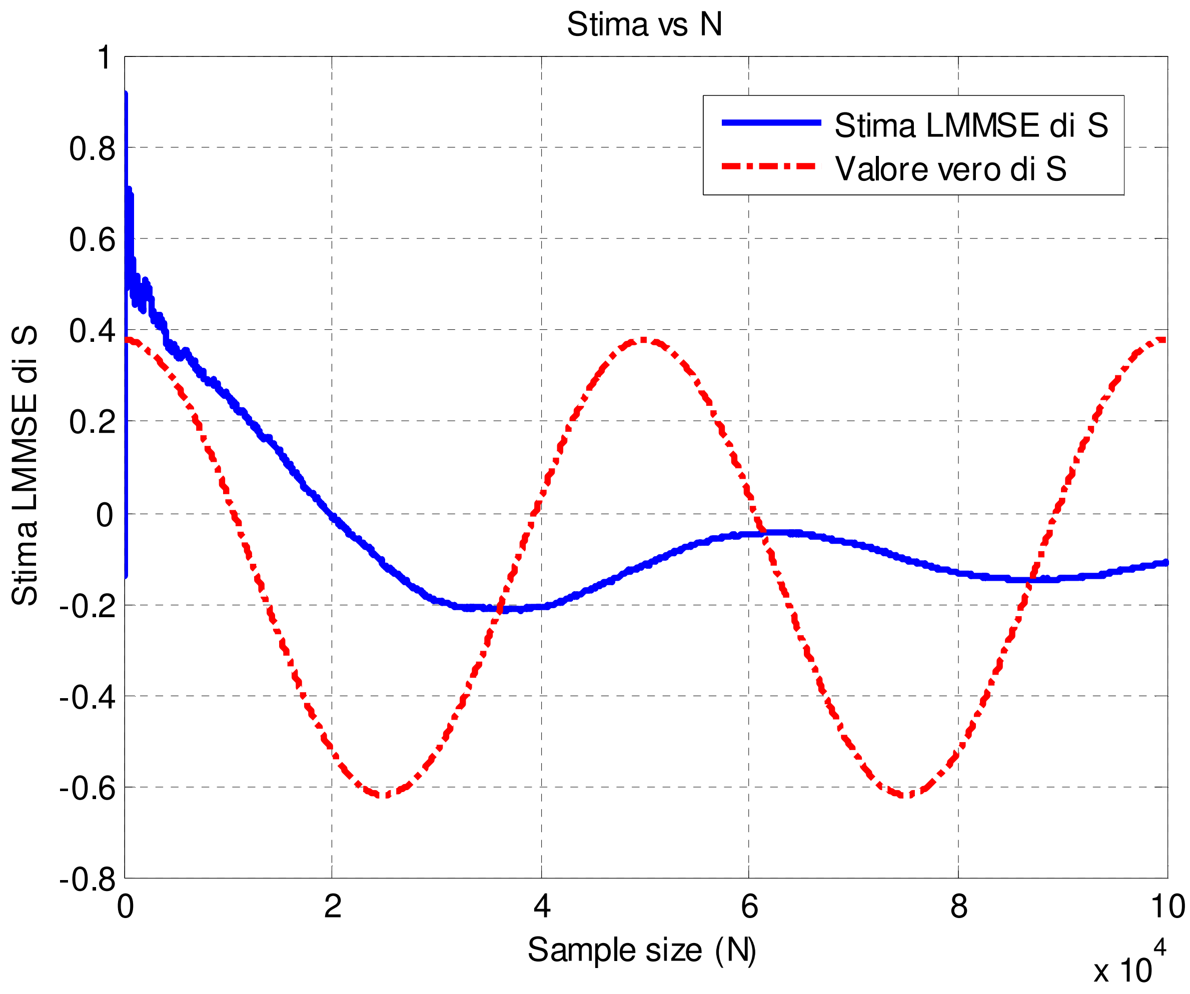
Sequential (Recursive) LMMSE Estimator

- If the amplitude of the variations of the SOI is larger, the low-pass effect is more evident.



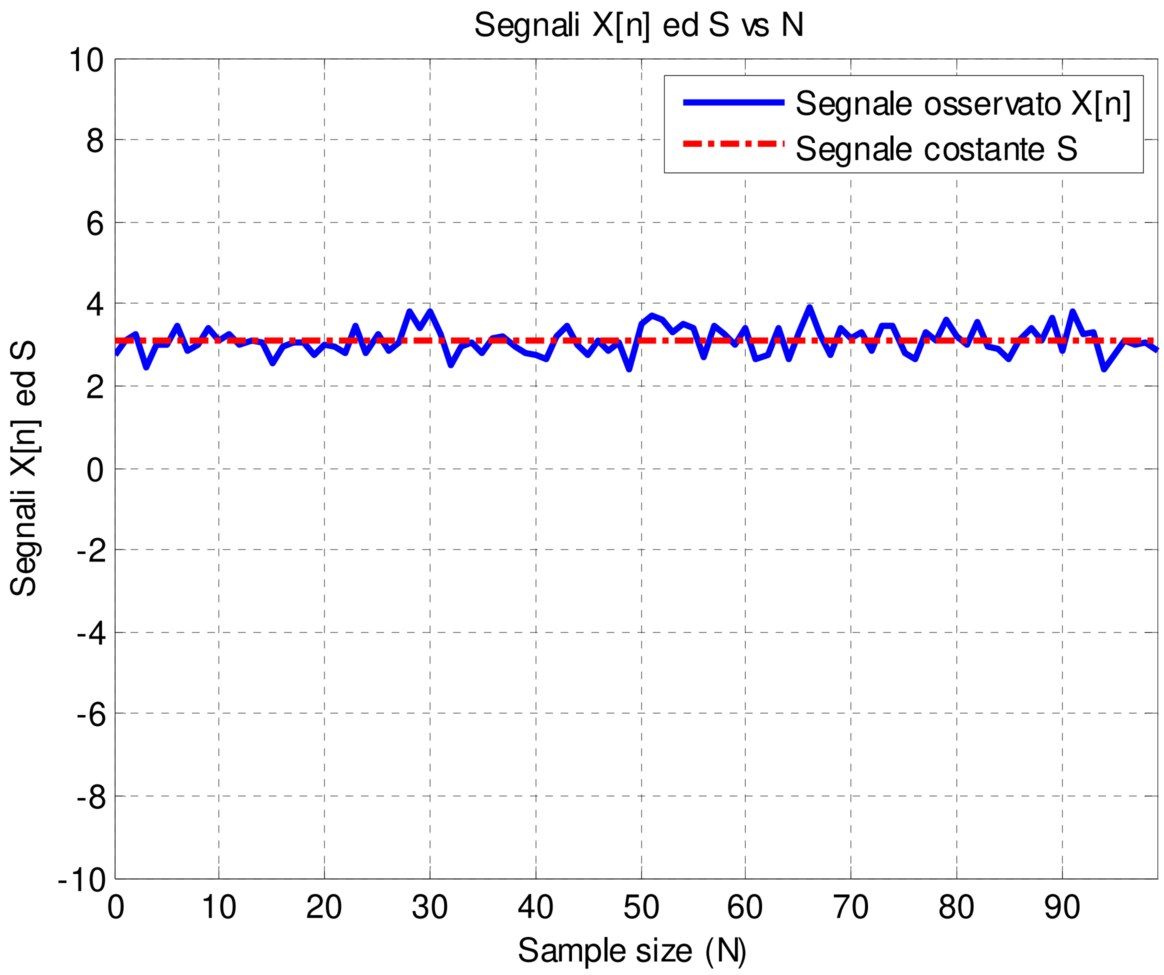
Sequential (Recursive) LMMSE Estimator

- If the amplitude of the variations of the SOI is larger, the low-pass effect is more evident.

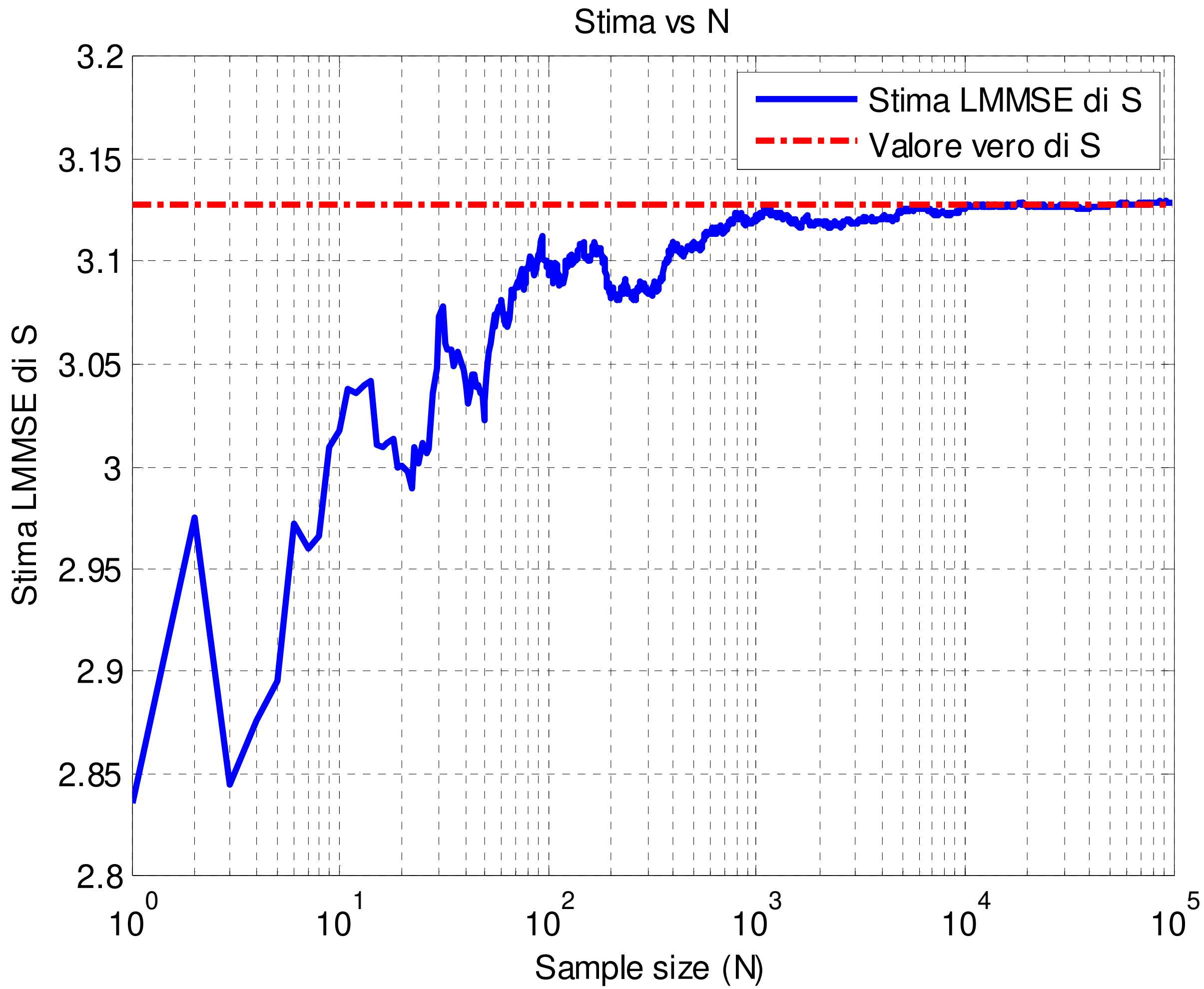


Sequential (Recursive) LMMSE Estimator

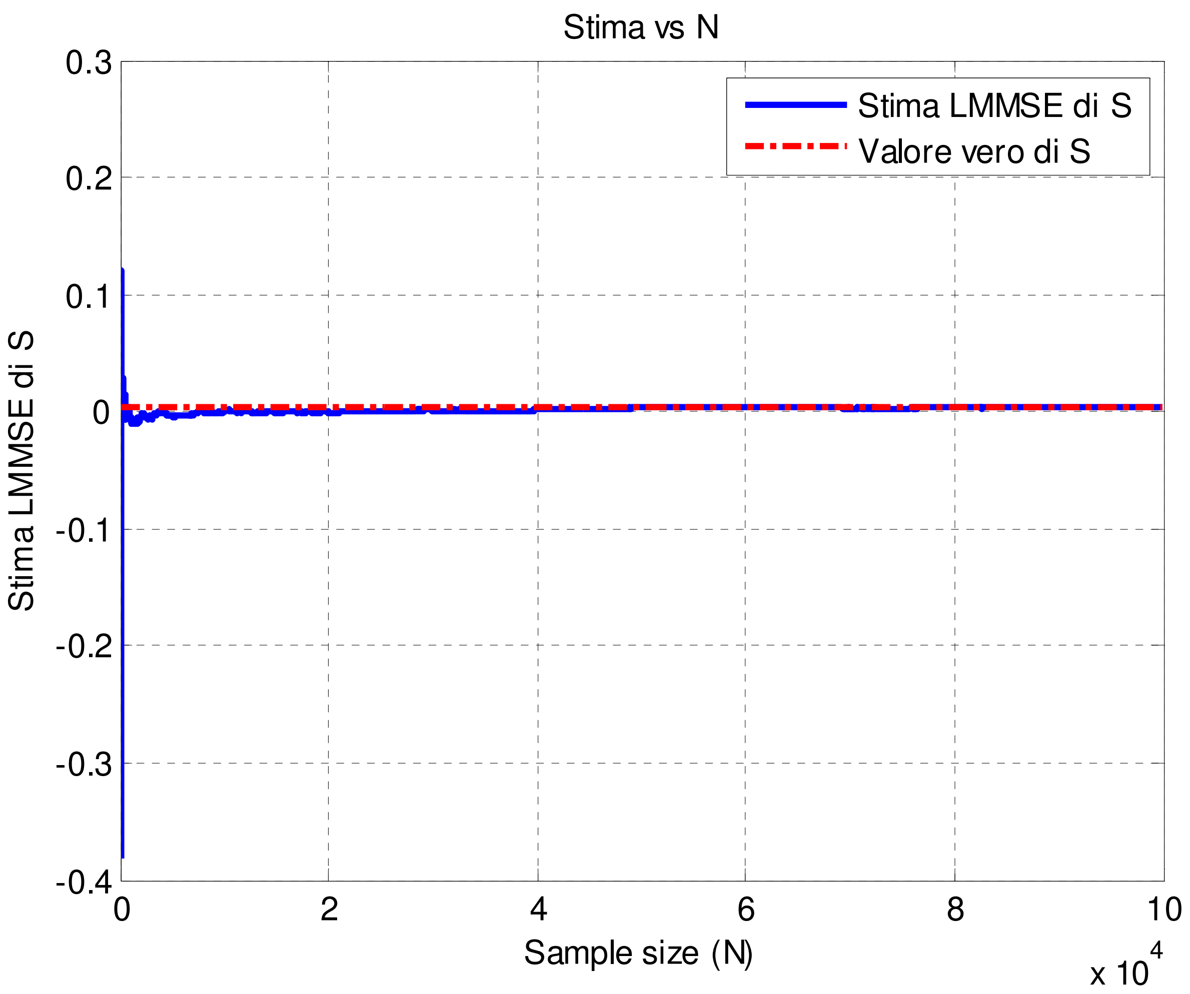
- Let us increase the *SNR*: $\gamma_{dB} = 10dB$

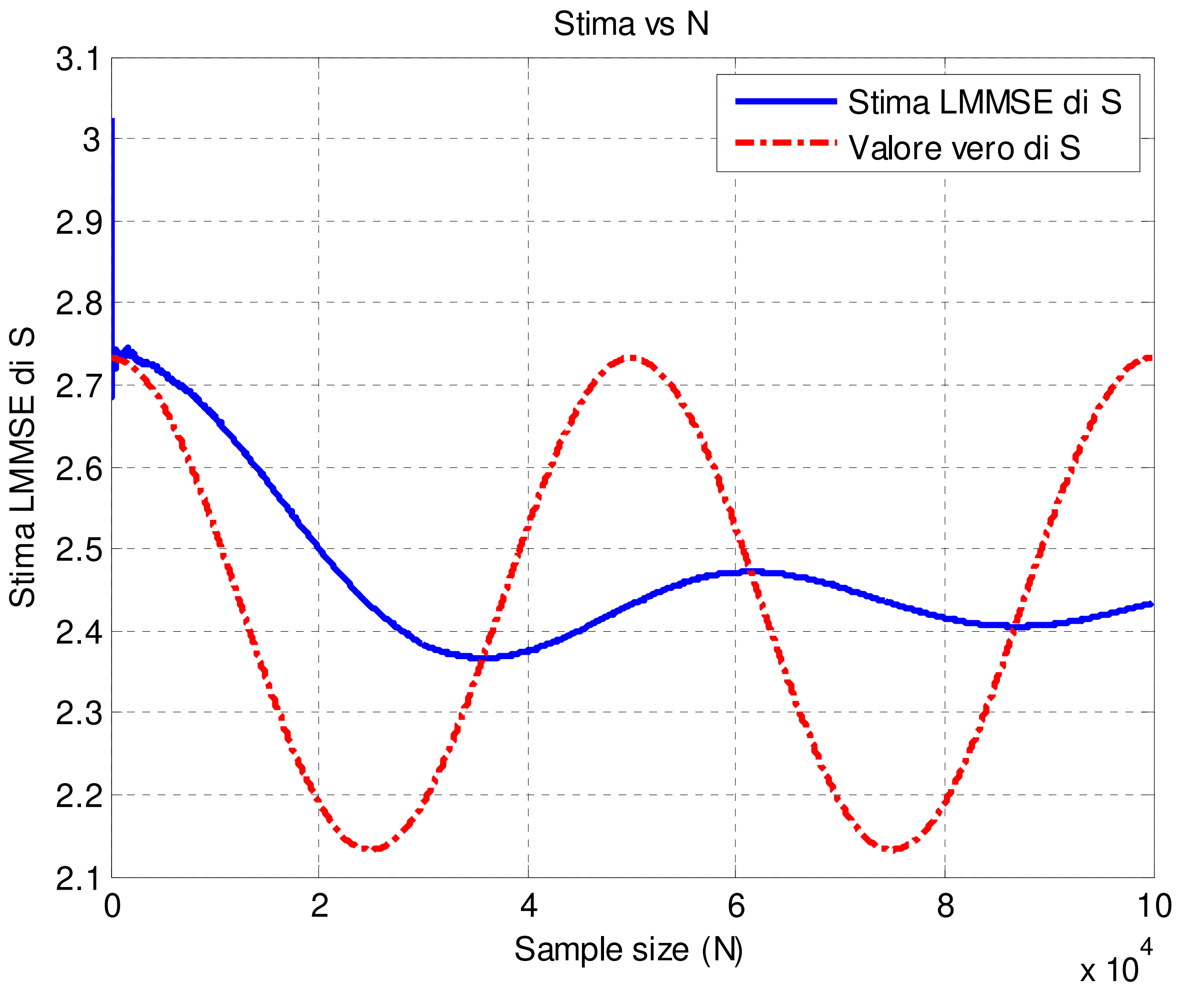


Sequential (Recursive) LMMSE Estimator



Sequential (Recursive) LMMSE Estimator





Stima vs N
Stima LMMSE di S
Valore vero di S
3.1
3
2.9
2.8
2.7
2.6
2.5
2.4
2.3
2.2
2.1
Stima LMMSE di S
0
2
4
6
8
10
Sample size (N)
x 10^4